

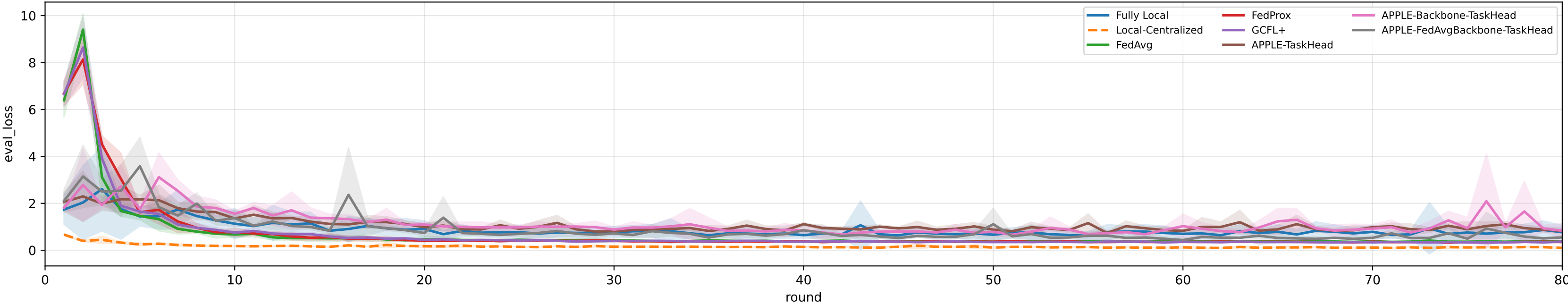
Controlled q-label heterogeneity: iid | aggregated / pooled performance over 5 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.25 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

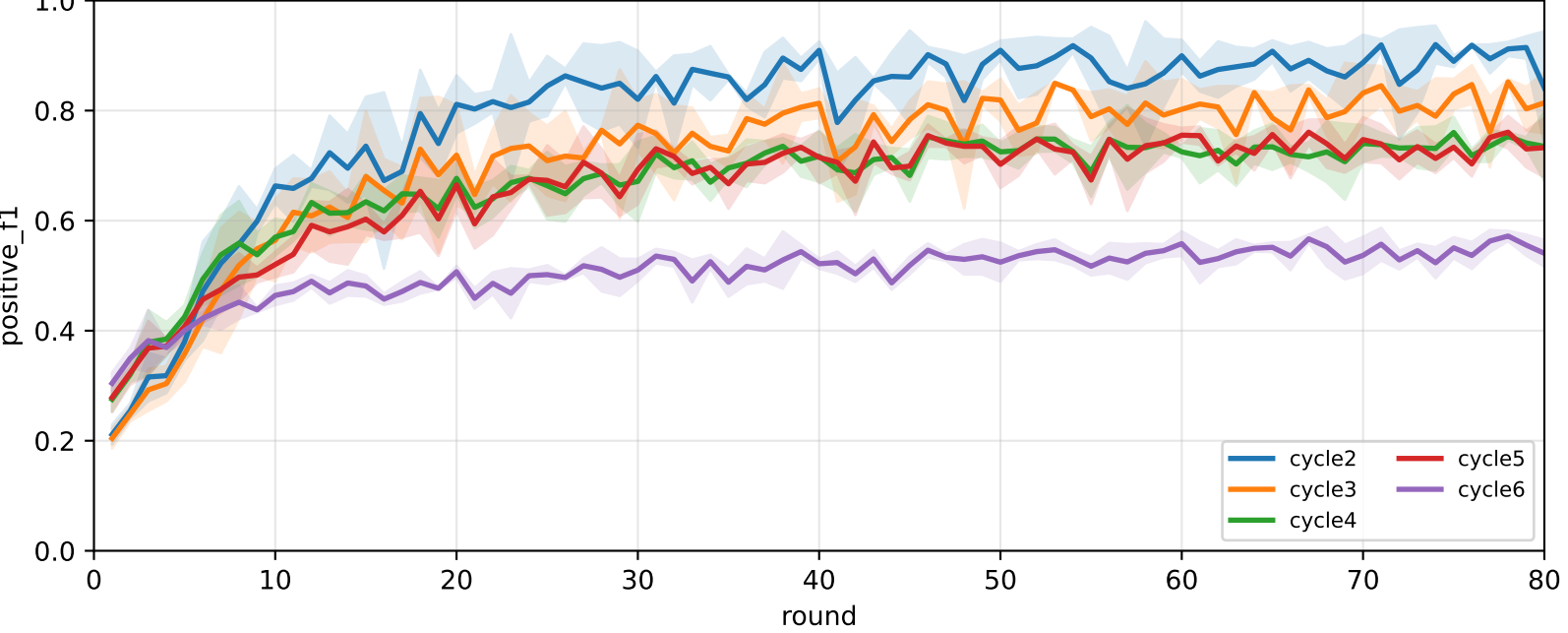
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	67.9% ± 1.6%	72.3% ± 1.5%	90.0% ± 1.5%	<u>85.0% ± 1.7%</u>	67.5% ± 3.5%	66.8% ± 2.2%	50.9% ± 1.2%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedAvg	70.2% ± 1.6%	73.5% ± 1.4%	91.2% ± 2.1%	82.8% ± 3.6%	72.4% ± 1.2%	64.1% ± 1.0%	56.9% ± 2.0%
FedProx	<u>71.9% ± 1.0%</u>	<u>75.0% ± 1.3%</u>	89.8% ± 3.9%	84.7% ± 1.9%	<u>76.3% ± 1.7%</u>	66.3% ± 1.0%	57.9% ± 0.2%
GCFL+	71.2% ± 1.2%	74.6% ± 0.9%	<u>92.5% ± 2.3%</u>	82.9% ± 3.1%	74.8% ± 1.1%	65.3% ± 1.4%	57.2% ± 1.2%
APPLE-TaskHead	67.4% ± 3.8%	70.0% ± 3.5%	85.0% ± 2.0%	75.6% ± 4.1%	67.3% ± 5.2%	65.0% ± 5.1%	54.8% ± 3.1%
APPLE-Backbone-TaskHead	64.3% ± 3.1%	66.0% ± 3.6%	72.1% ± 10.7%	71.3% ± 3.2%	67.5% ± 3.3%	62.6% ± 3.7%	54.7% ± 1.5%
APPLE-FedAvgBackbone-TaskHead	71.1% ± 0.4%	74.0% ± 0.9%	88.2% ± 2.6%	82.6% ± 0.7%	69.4% ± 3.6%	<u>70.0% ± 1.6%</u>	<u>58.1% ± 1.3%</u>

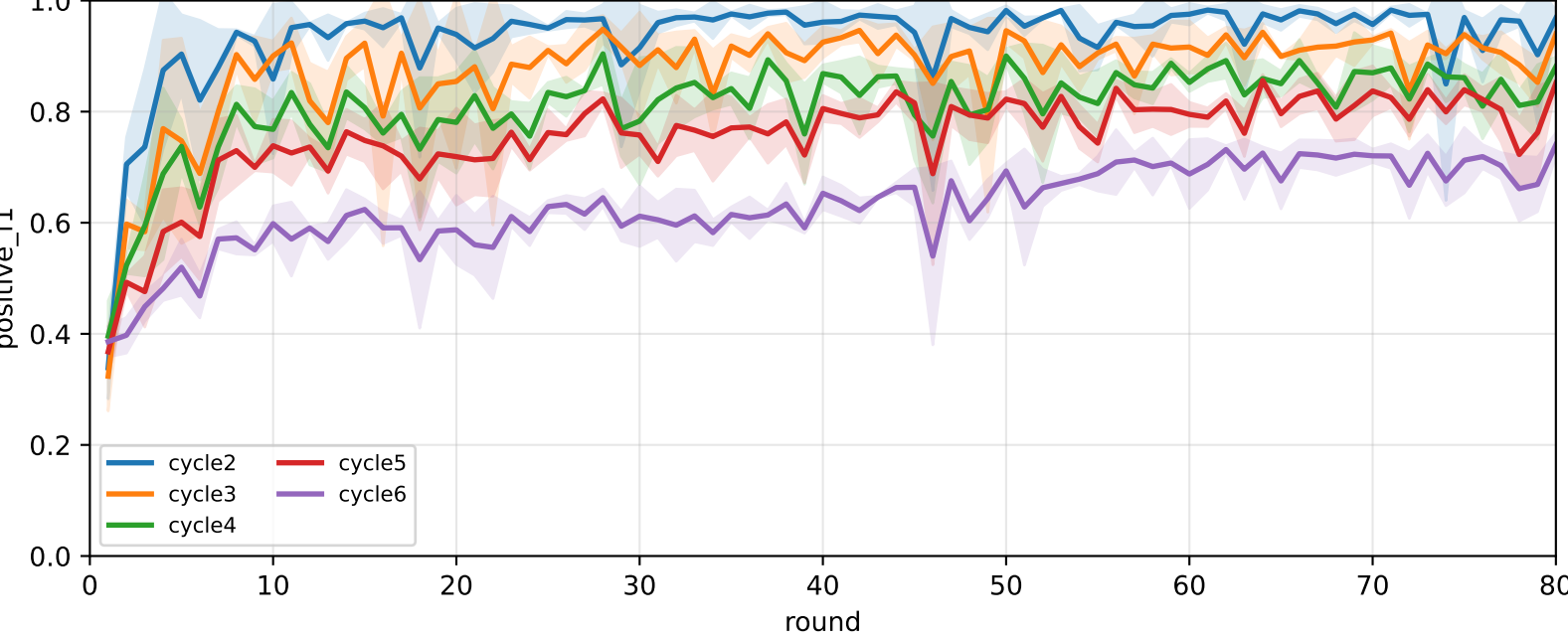
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



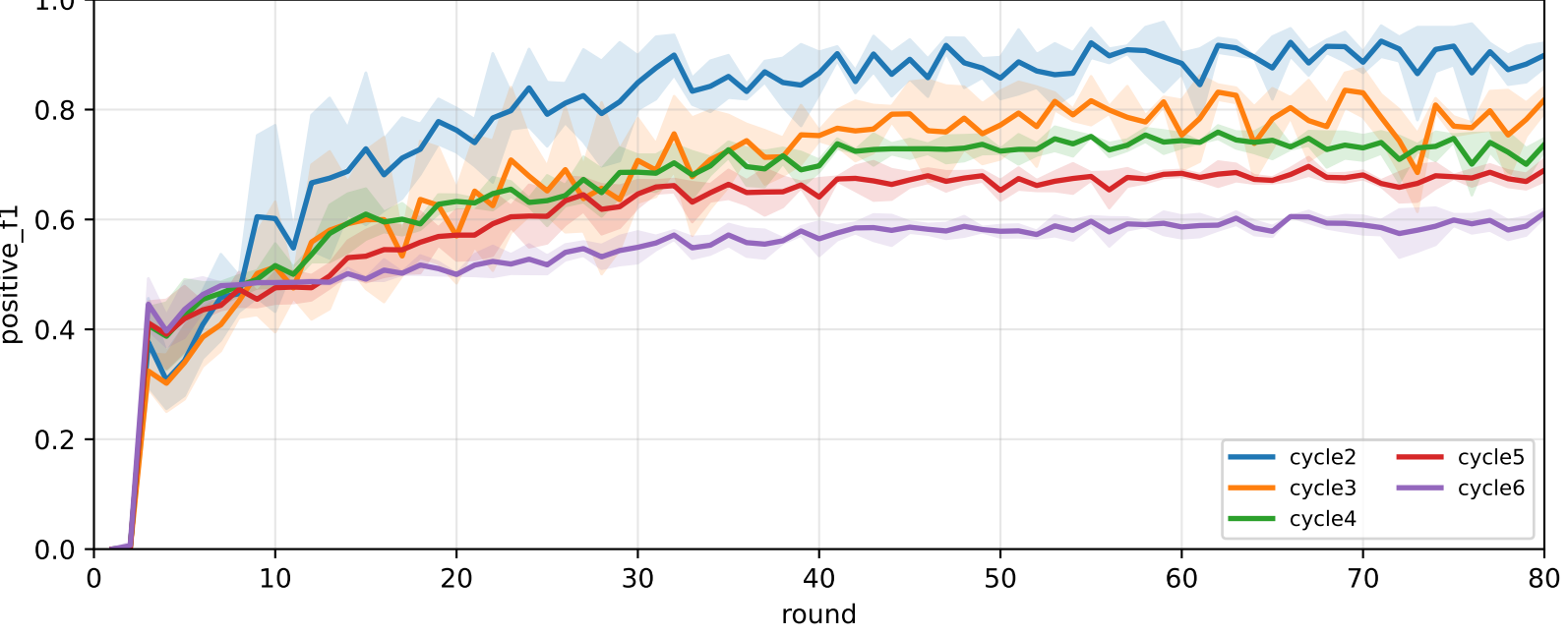
Pooled Fully Local positive F1



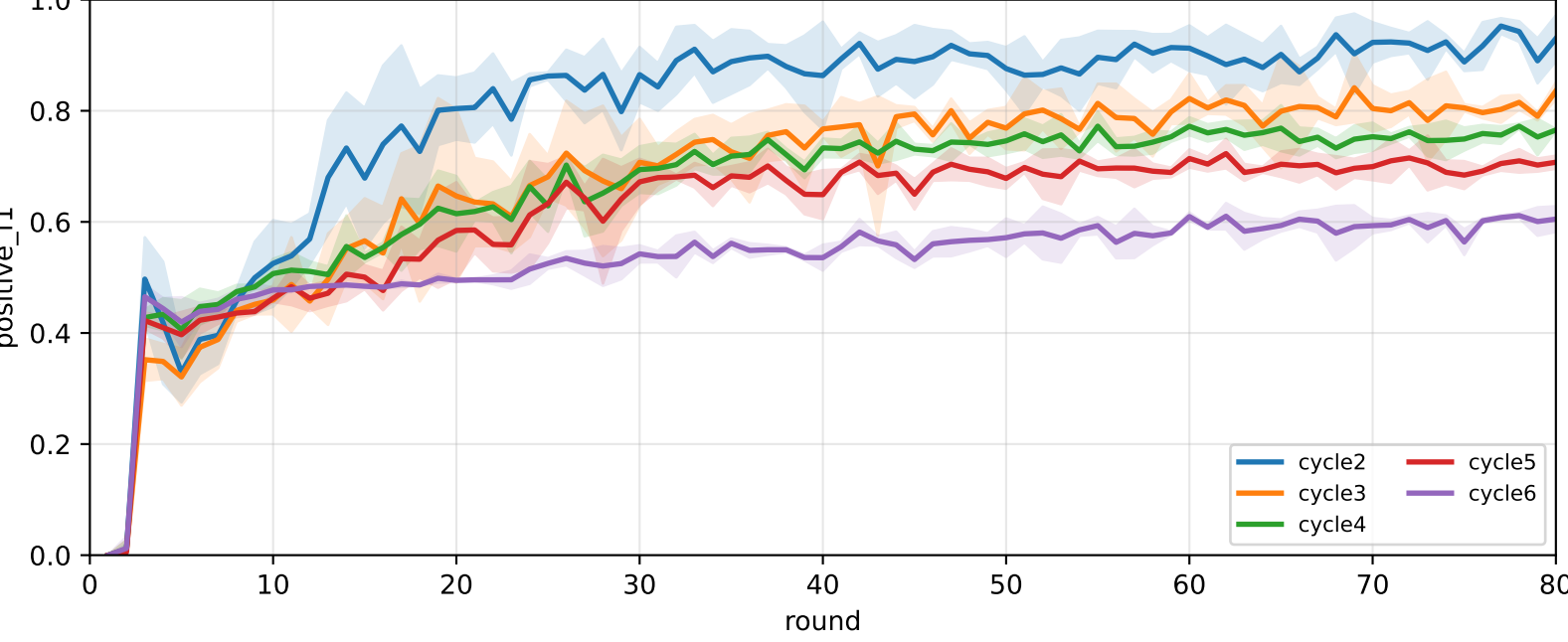
Pooled Local-Centralized positive F1



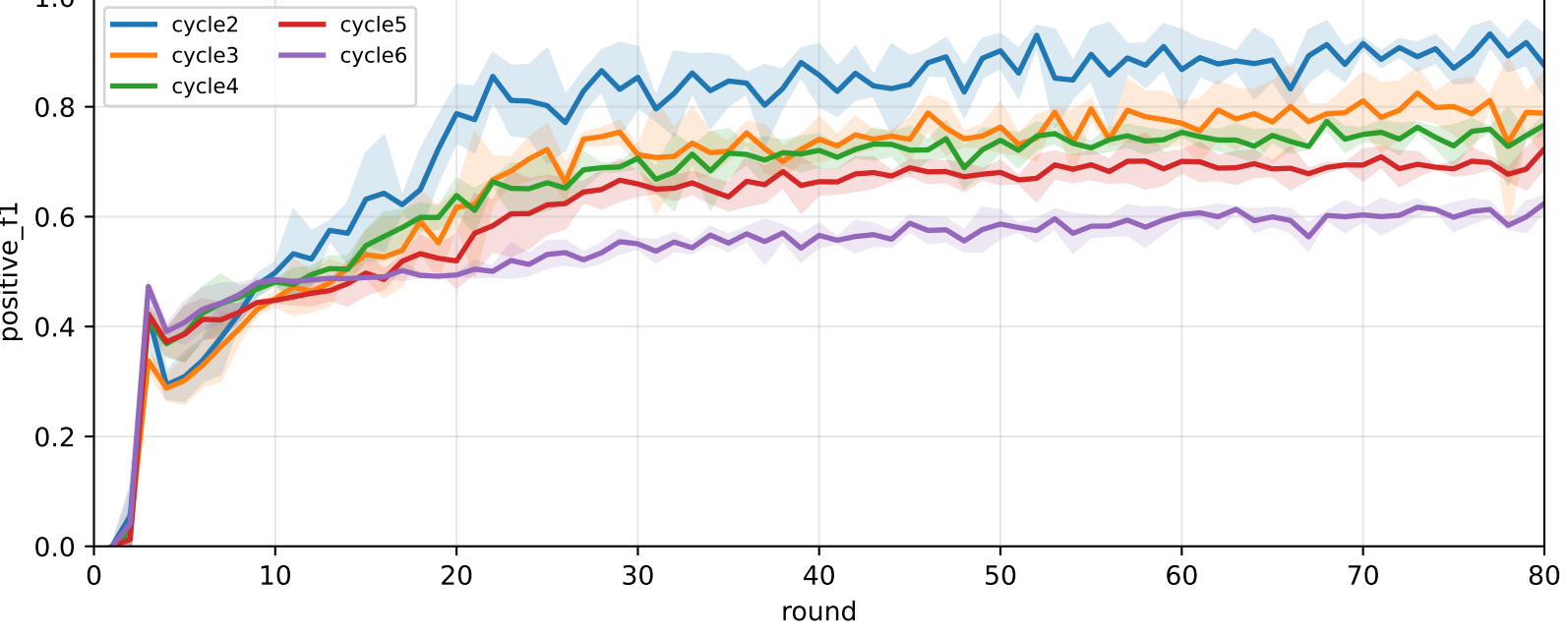
Pooled FedAvg positive F1



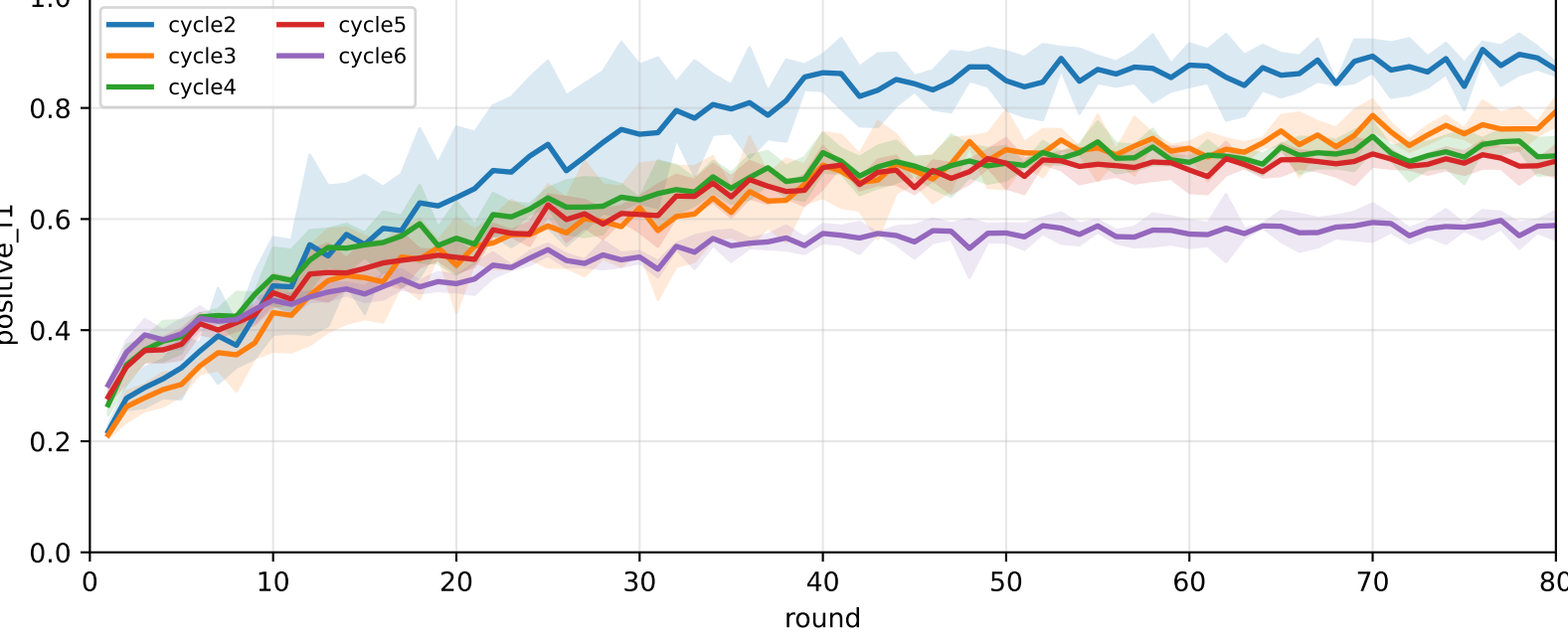
Pooled FedProx positive F1



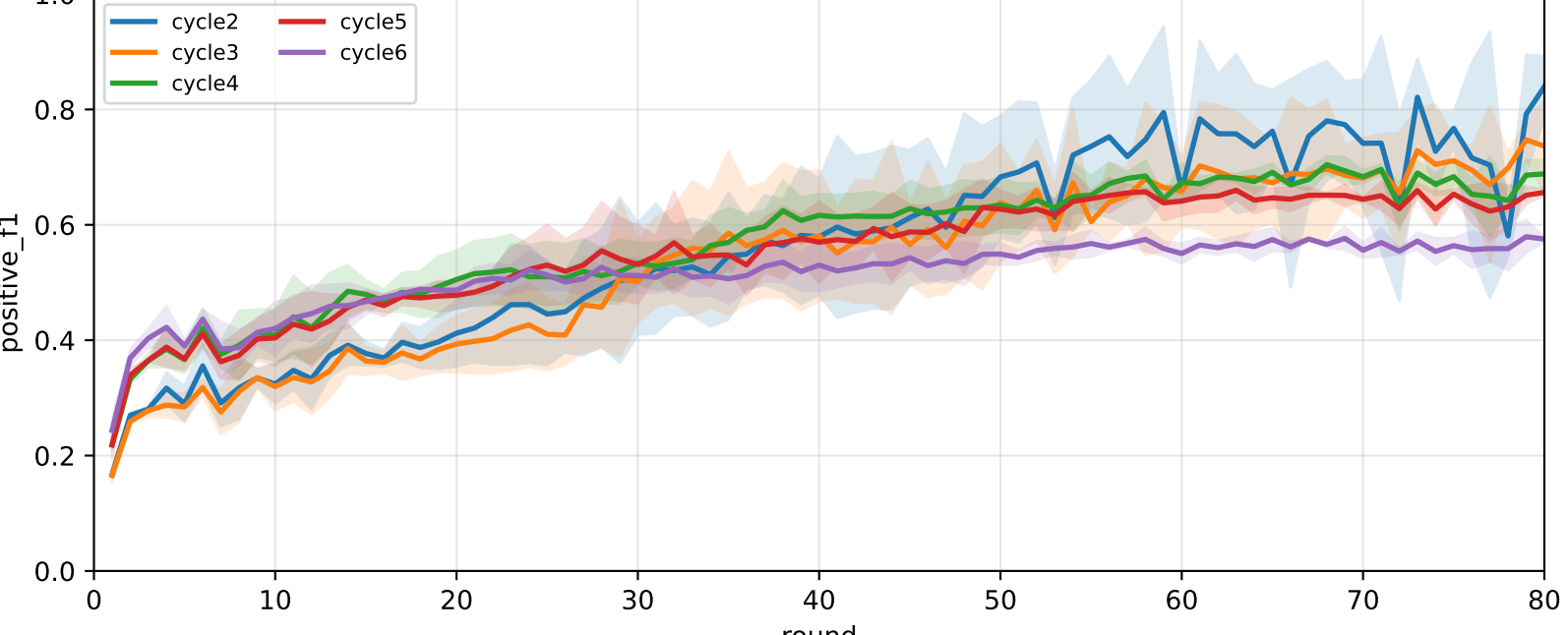
Pooled GCFL+ positive F1



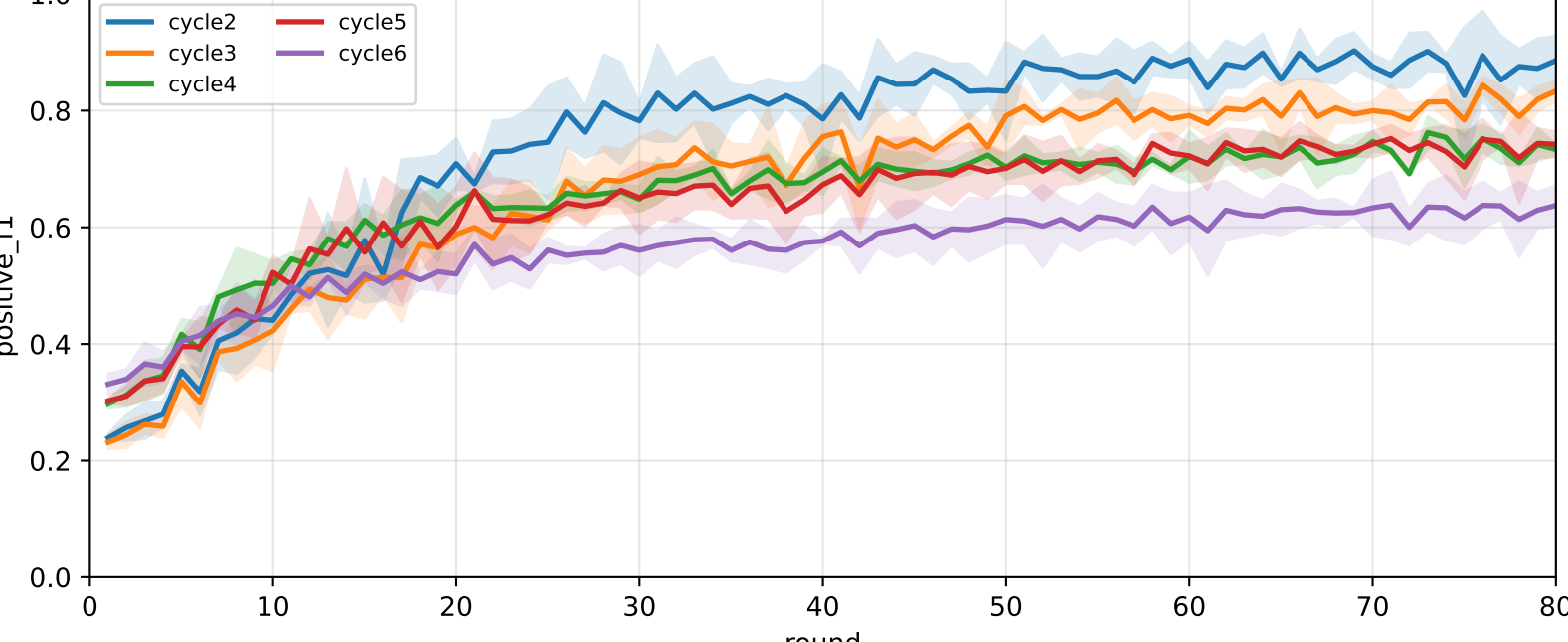
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)

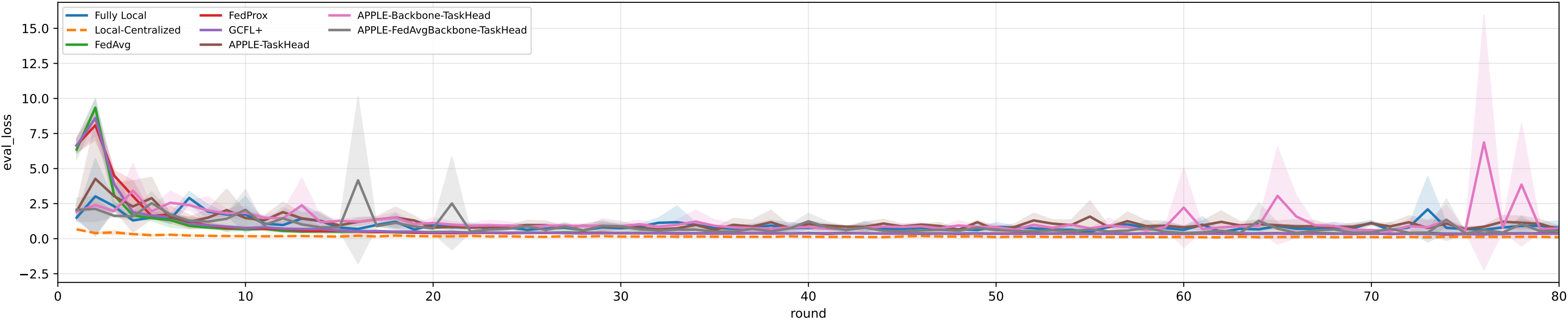


family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

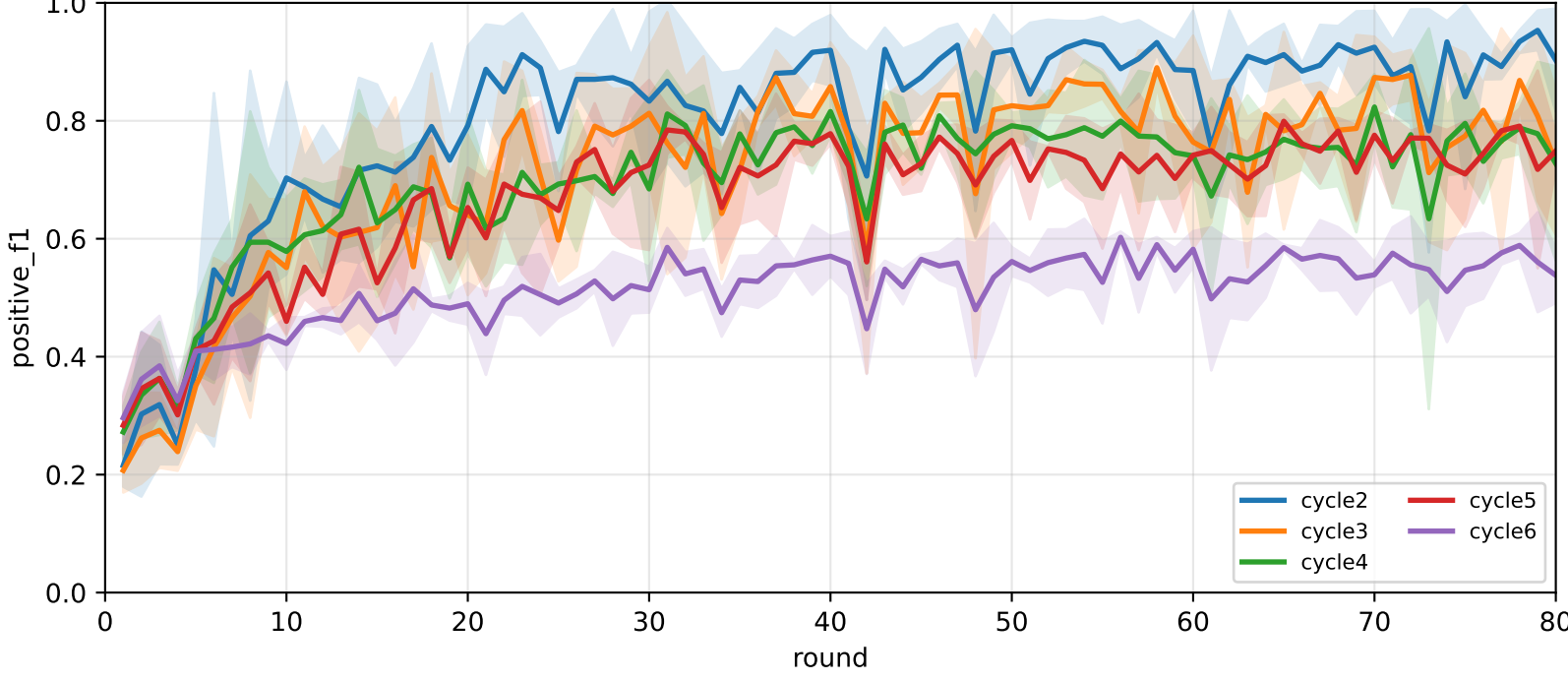
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	70.1% ± 3.2%	74.3% ± 2.5%	93.1% ± 1.1%	<u>86.7% ± 1.6%</u>	70.3% ± 4.6%	<u>67.5% ± 2.2%</u>	53.8% ± 5.0%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedAvg	70.3% ± 2.1%	73.5% ± 2.0%	90.3% ± 3.0%	82.8% ± 5.0%	72.6% ± 1.9%	64.5% ± 1.4%	57.1% ± 2.4%
FedProx	<u>71.6% ± 0.8%</u>	74.6% ± 1.1%	88.8% ± 4.3%	84.0% ± 1.4%	<u>76.1% ± 1.4%</u>	66.0% ± 1.2%	<u>58.0% ± 0.0%</u>
GCFL+	71.4% ± 1.1%	<u>74.8% ± 0.7%</u>	<u>93.2% ± 3.0%</u>	82.6% ± 3.5%	75.4% ± 0.7%	65.3% ± 1.1%	57.3% ± 1.4%
APPLE-TaskHead	63.7% ± 5.0%	66.7% ± 5.2%	84.8% ± 3.6%	71.8% ± 10.1%	65.5% ± 6.7%	59.8% ± 3.6%	51.9% ± 4.4%
APPLE-Backbone-TaskHead	65.8% ± 4.0%	67.9% ± 4.7%	78.0% ± 13.8%	74.4% ± 7.5%	69.1% ± 3.6%	62.3% ± 3.5%	55.9% ± 2.6%
APPLE-FedAvgBackbone-TaskHead	67.3% ± 2.7%	71.2% ± 3.4%	86.5% ± 4.7%	85.5% ± 5.1%	66.2% ± 7.5%	64.9% ± 5.6%	53.1% ± 2.6%

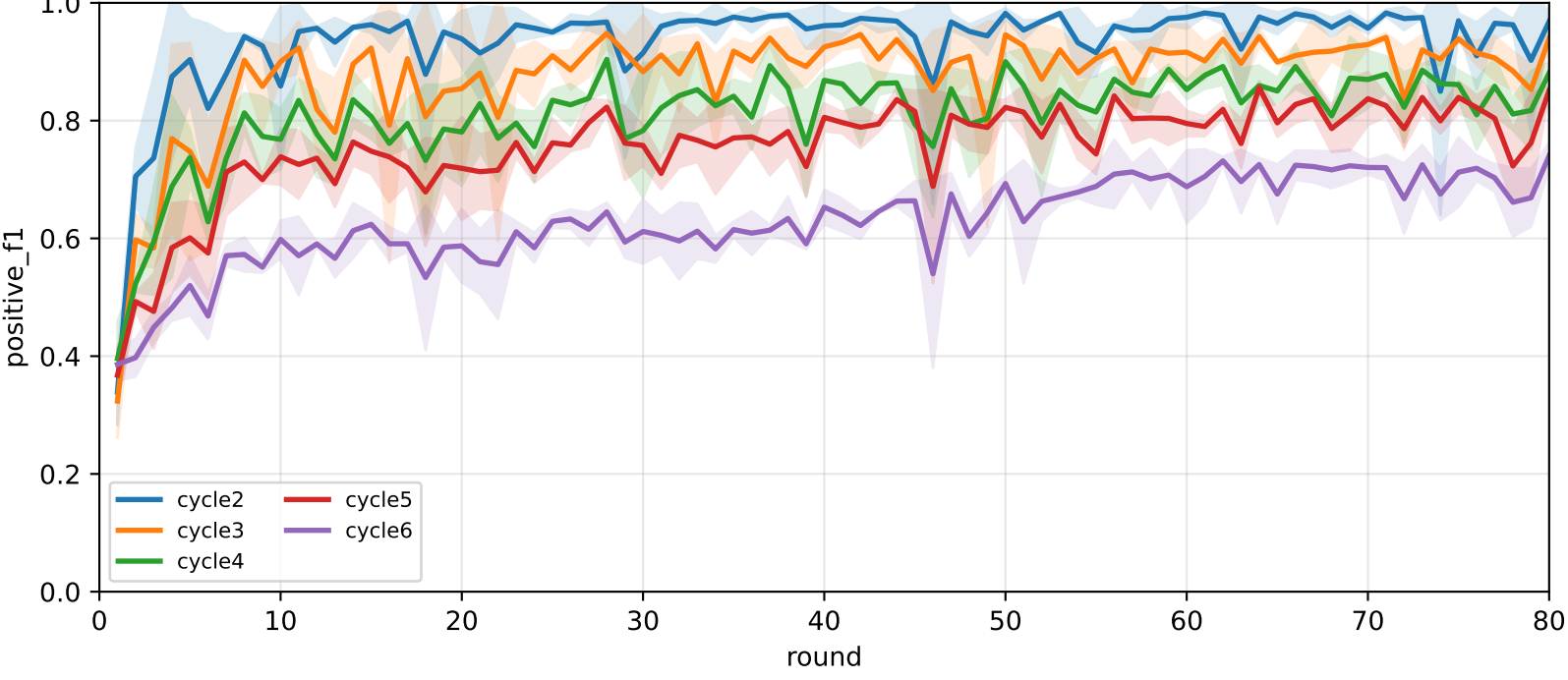
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



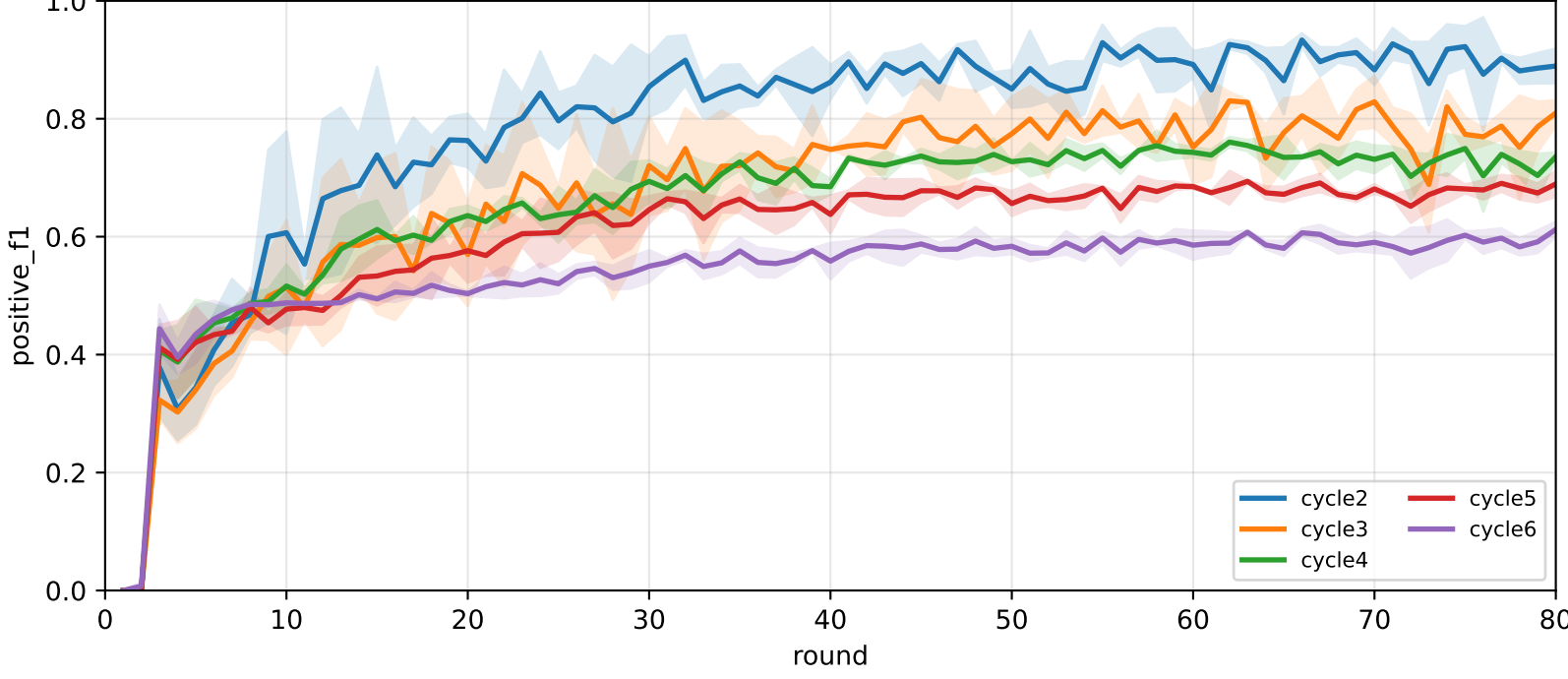
Pooled Fully Local positive F1



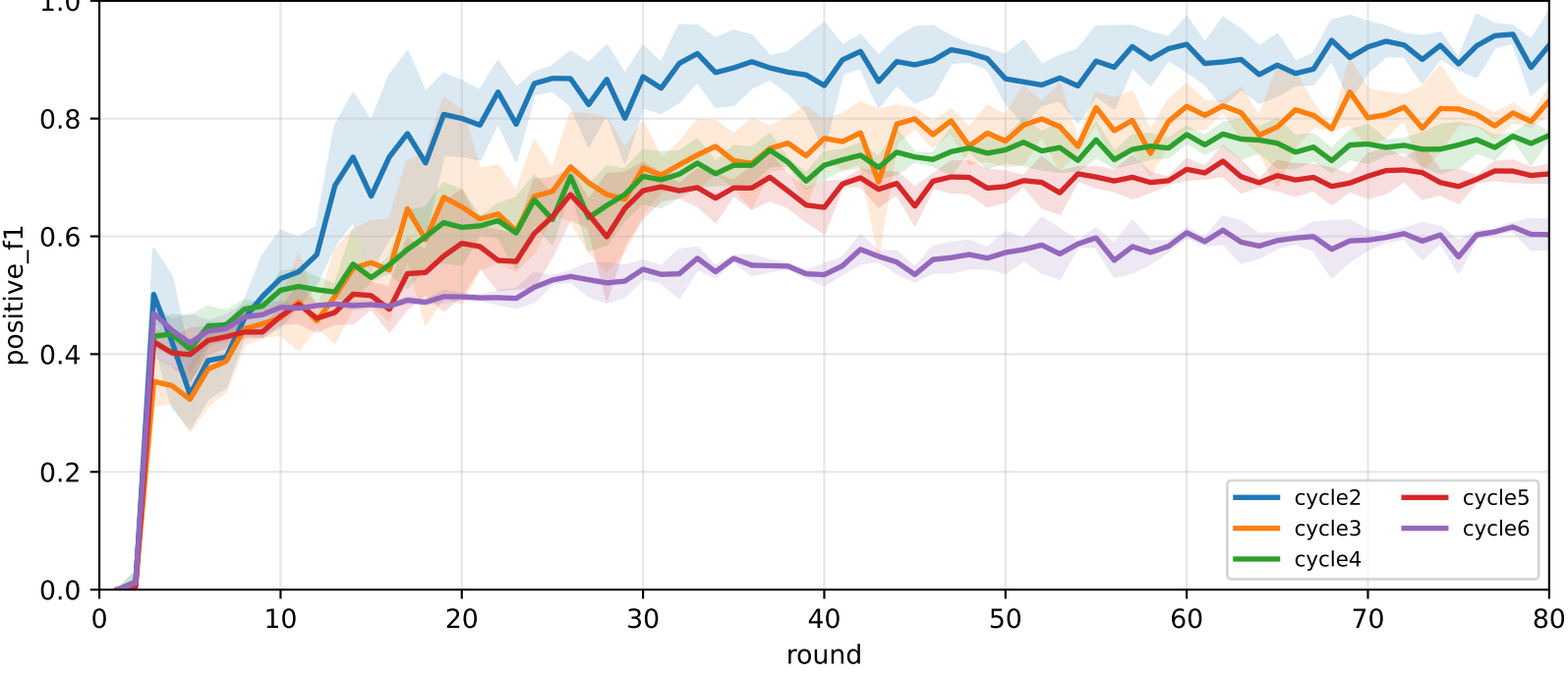
Pooled Local-Centralized positive F1



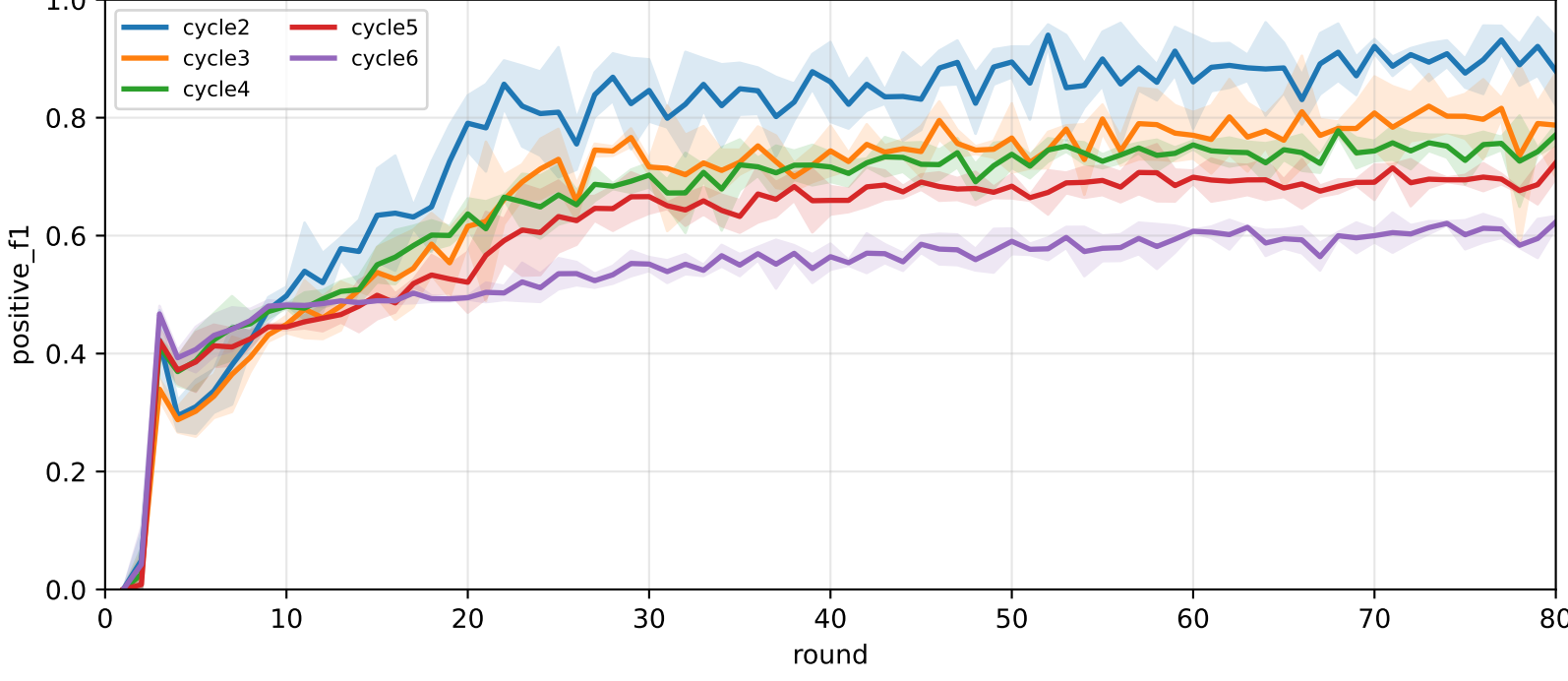
Pooled FedAvg positive F1



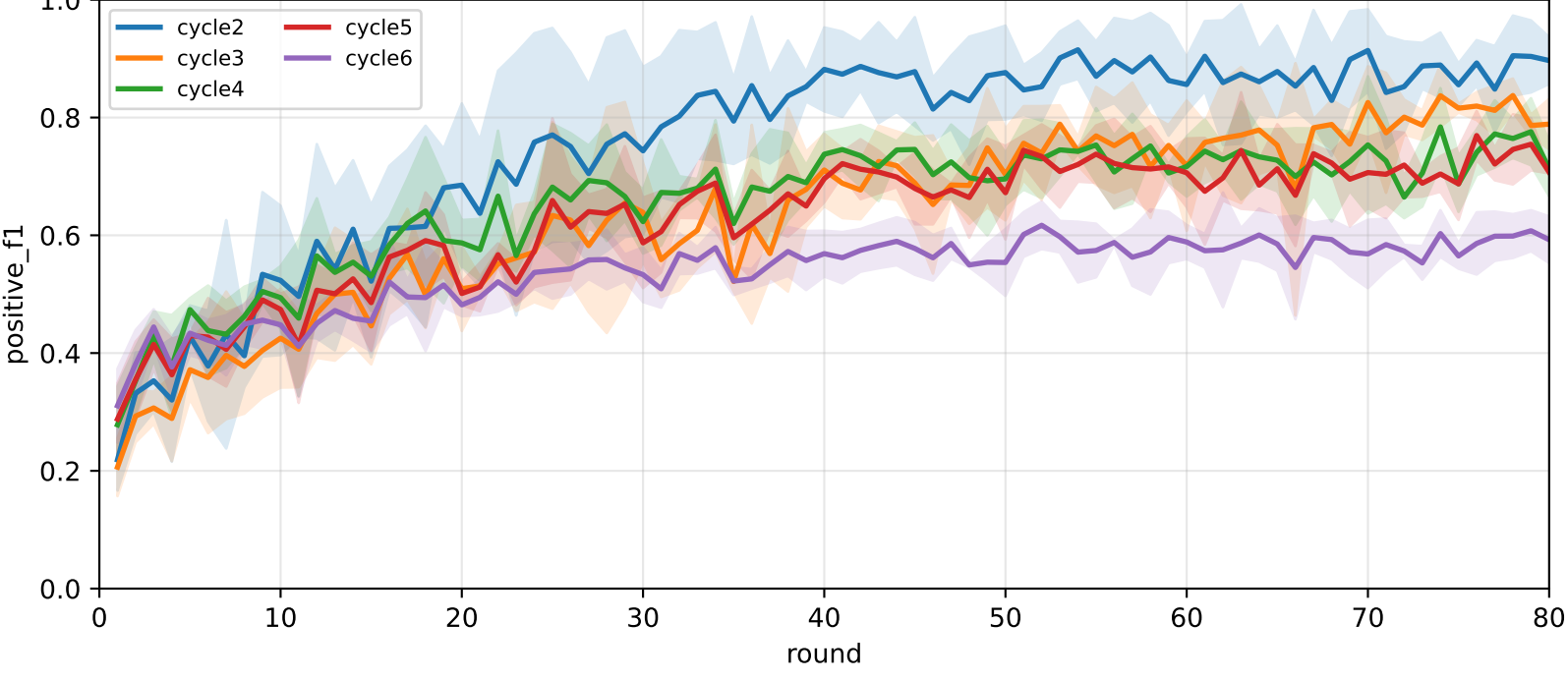
Pooled FedProx positive F1



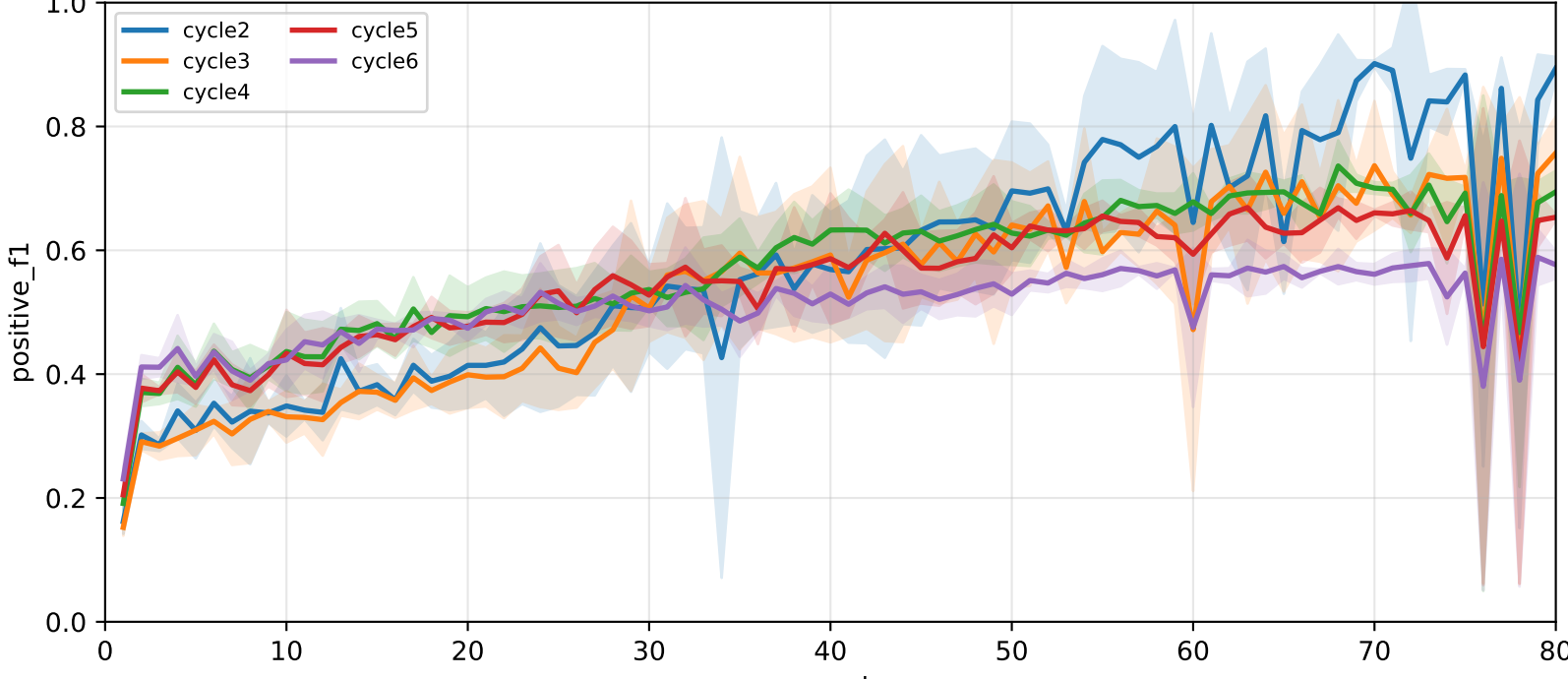
Pooled GCFL+ positive F1



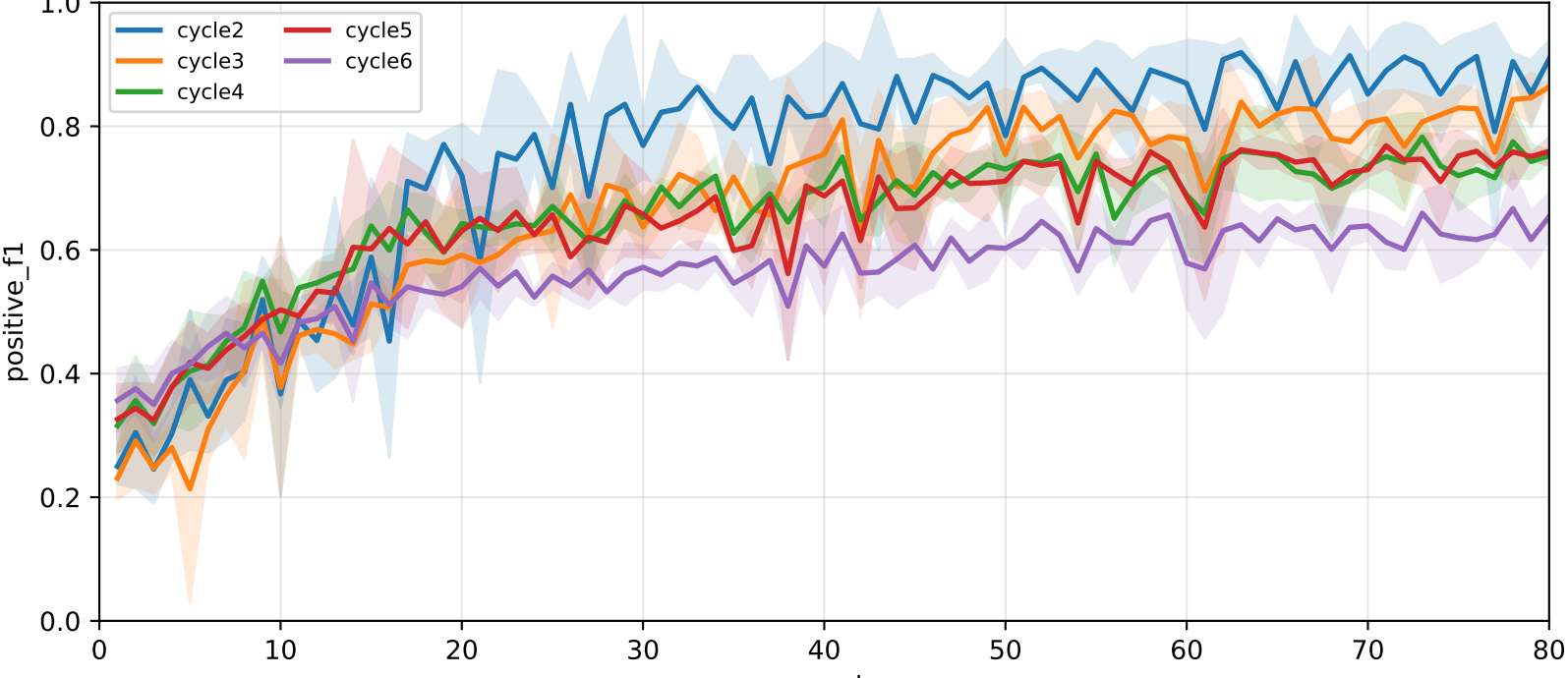
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



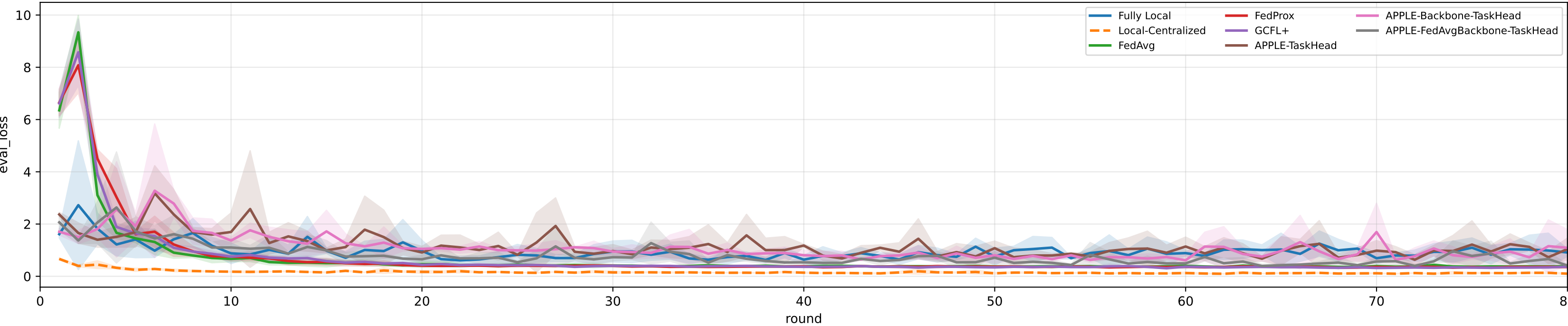
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q task_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

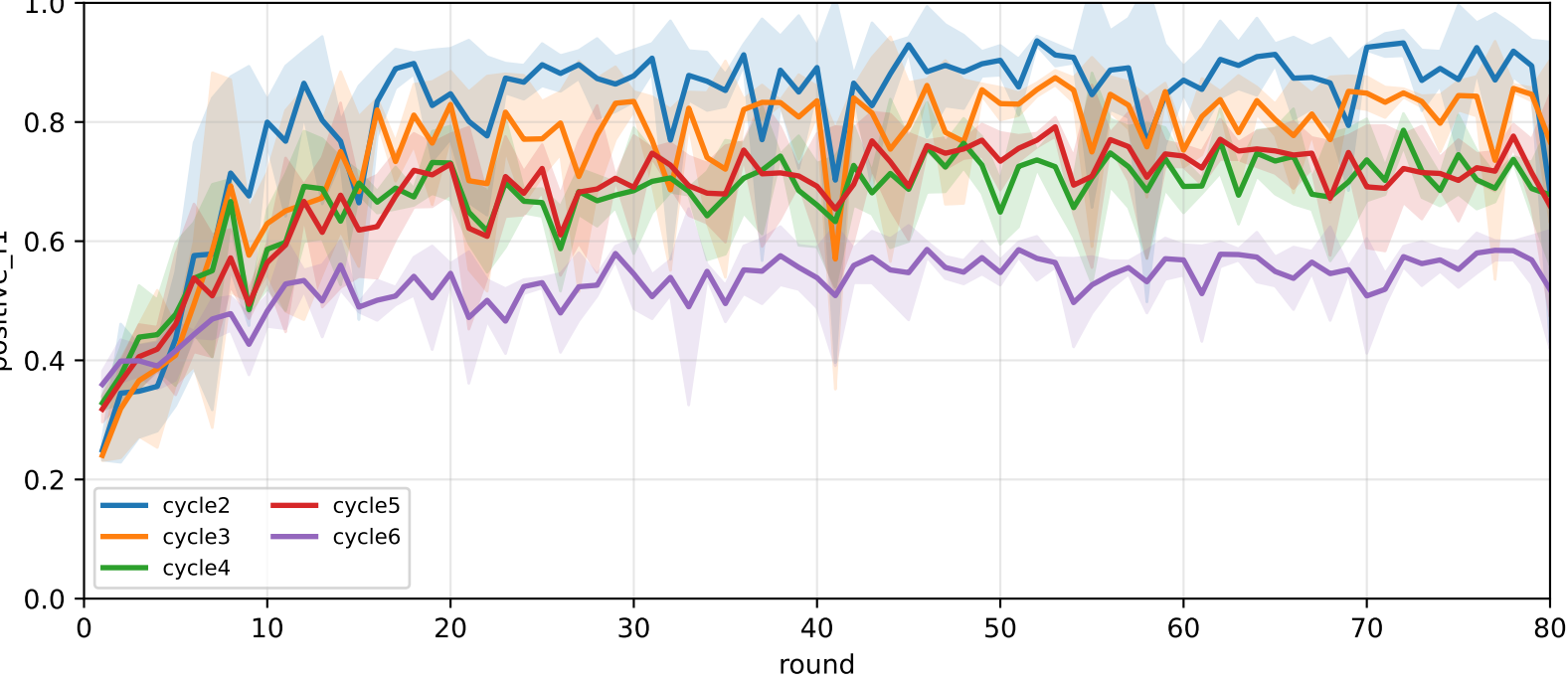
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	69.5% ± 0.8%	73.5% ± 0.6%	90.4% ± 0.9%	<u>85.2% ± 1.6%</u>	67.7% ± 0.6%	<u>71.2% ± 1.1%</u>	52.9% ± 2.1%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedAvg	70.5% ± 1.8%	73.9% ± 1.5%	<u>92.8% ± 1.8%</u>	82.1% ± 3.6%	73.3% ± 1.6%	64.4% ± 1.5%	56.9% ± 2.3%
FedProx	<u>71.9% ± 0.9%</u>	<u>75.1% ± 1.0%</u>	90.3% ± 3.6%	84.6% ± 1.7%	<u>76.2% ± 2.0%</u>	66.3% ± 1.1%	57.9% ± 0.6%
GCFL+	71.2% ± 1.7%	74.5% ± 1.5%	91.9% ± 1.0%	83.2% ± 3.5%	74.0% ± 2.0%	66.3% ± 2.1%	57.1% ± 1.1%
APPLE-TaskHead	66.7% ± 5.8%	69.6% ± 5.5%	84.3% ± 5.8%	81.0% ± 3.7%	65.1% ± 7.6%	64.5% ± 8.0%	53.2% ± 3.8%
APPLE-Backbone-TaskHead	65.4% ± 2.5%	67.6% ± 3.2%	76.8% ± 15.1%	73.7% ± 6.1%	67.2% ± 2.9%	64.9% ± 1.7%	55.2% ± 1.3%
APPLE-FedAvgBackbone-TaskHead	71.3% ± 2.0%	74.3% ± 1.6%	91.2% ± 1.6%	84.7% ± 0.6%	65.8% ± 5.9%	70.3% ± 3.1%	<u>59.7% ± 2.2%</u>

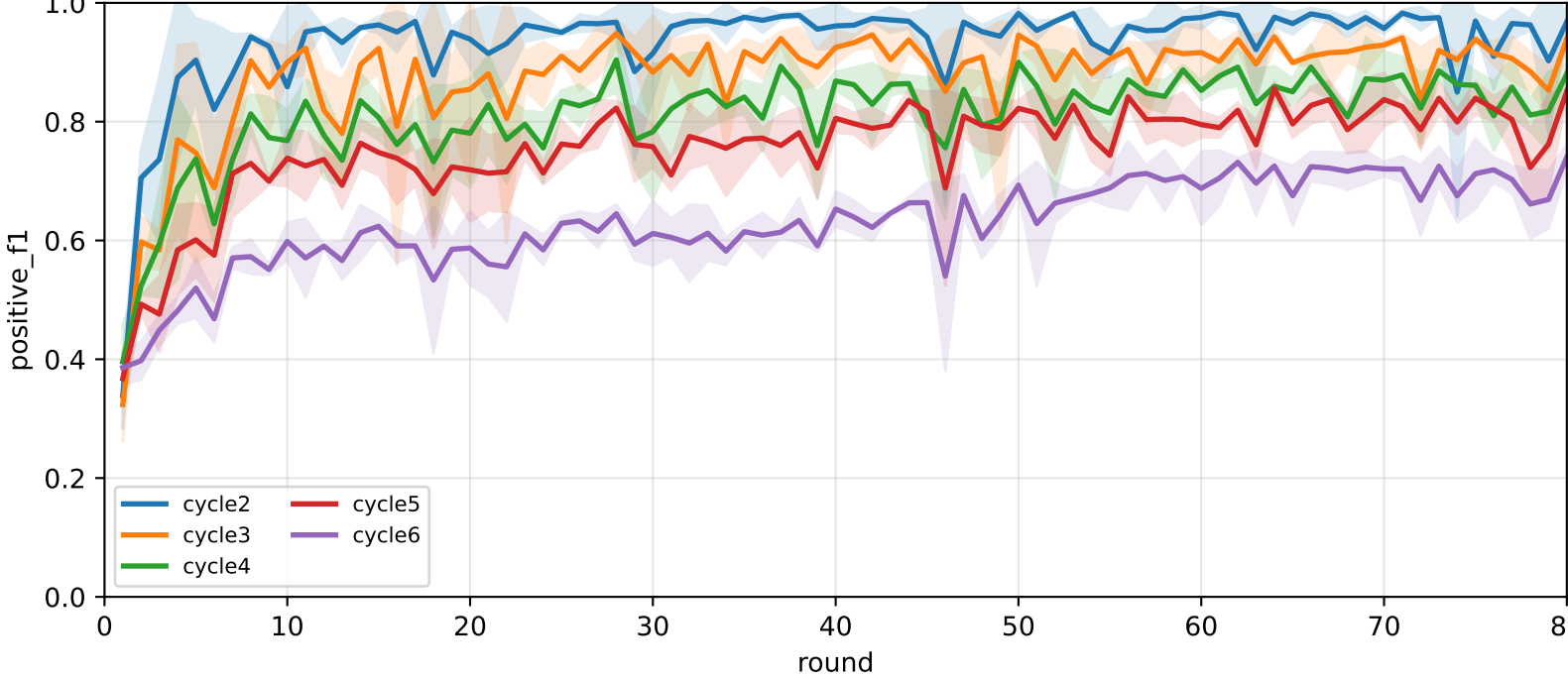
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



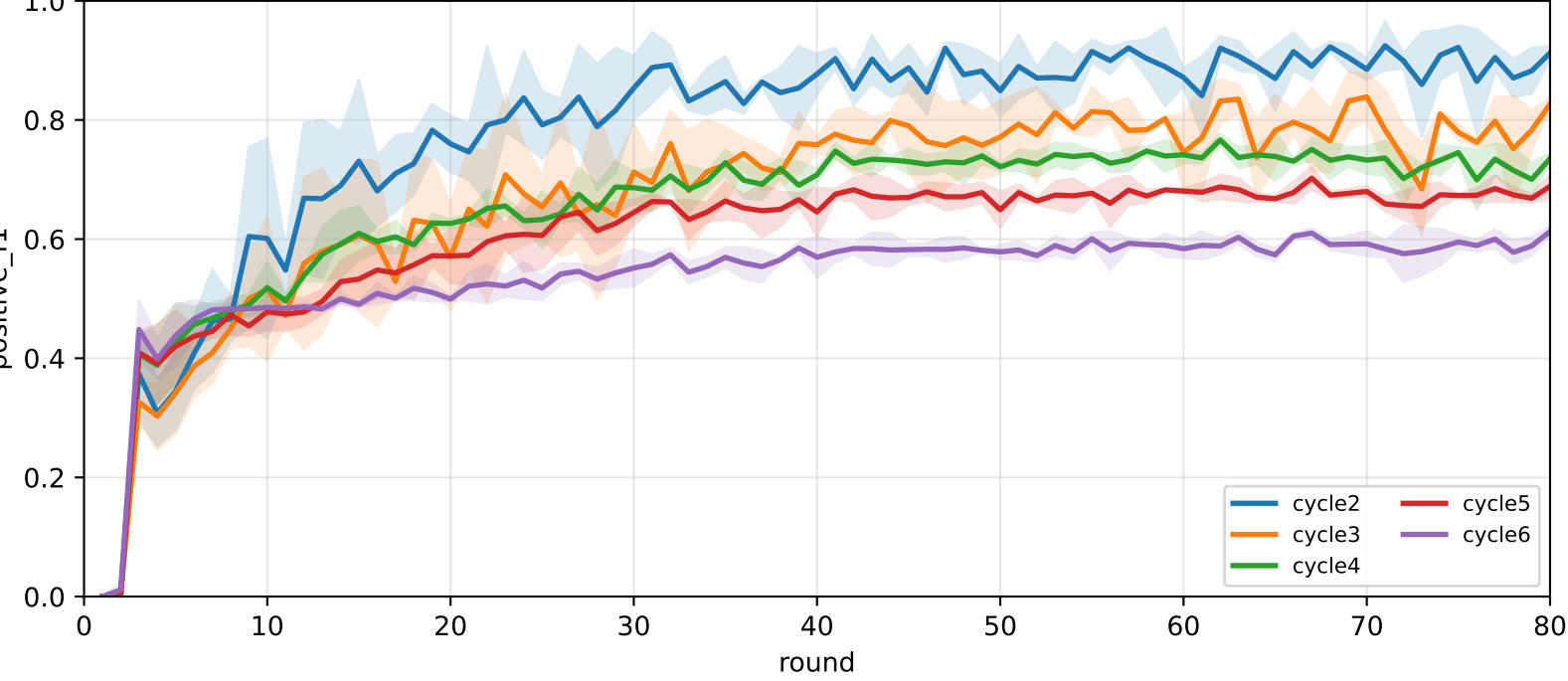
Pooled Fully Local positive F1



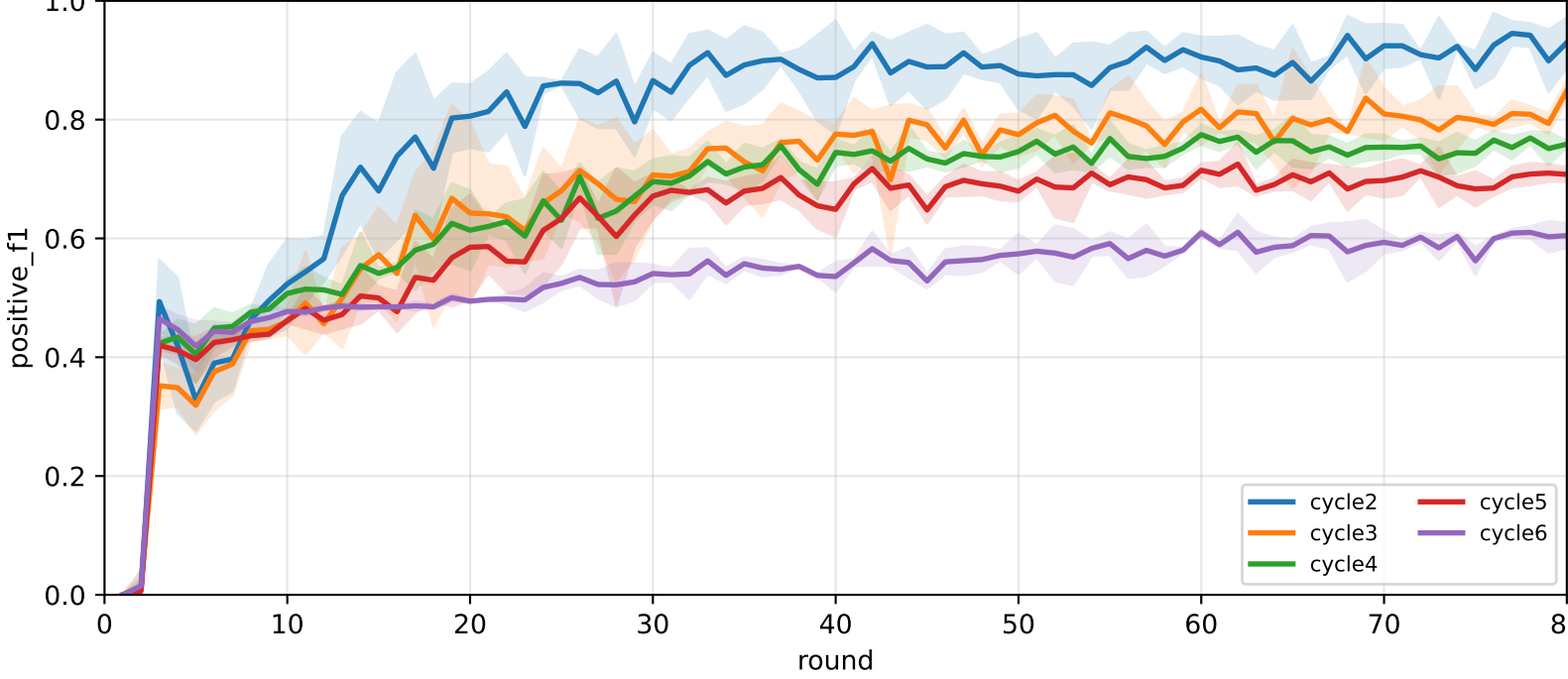
Pooled Local-Centralized positive F1



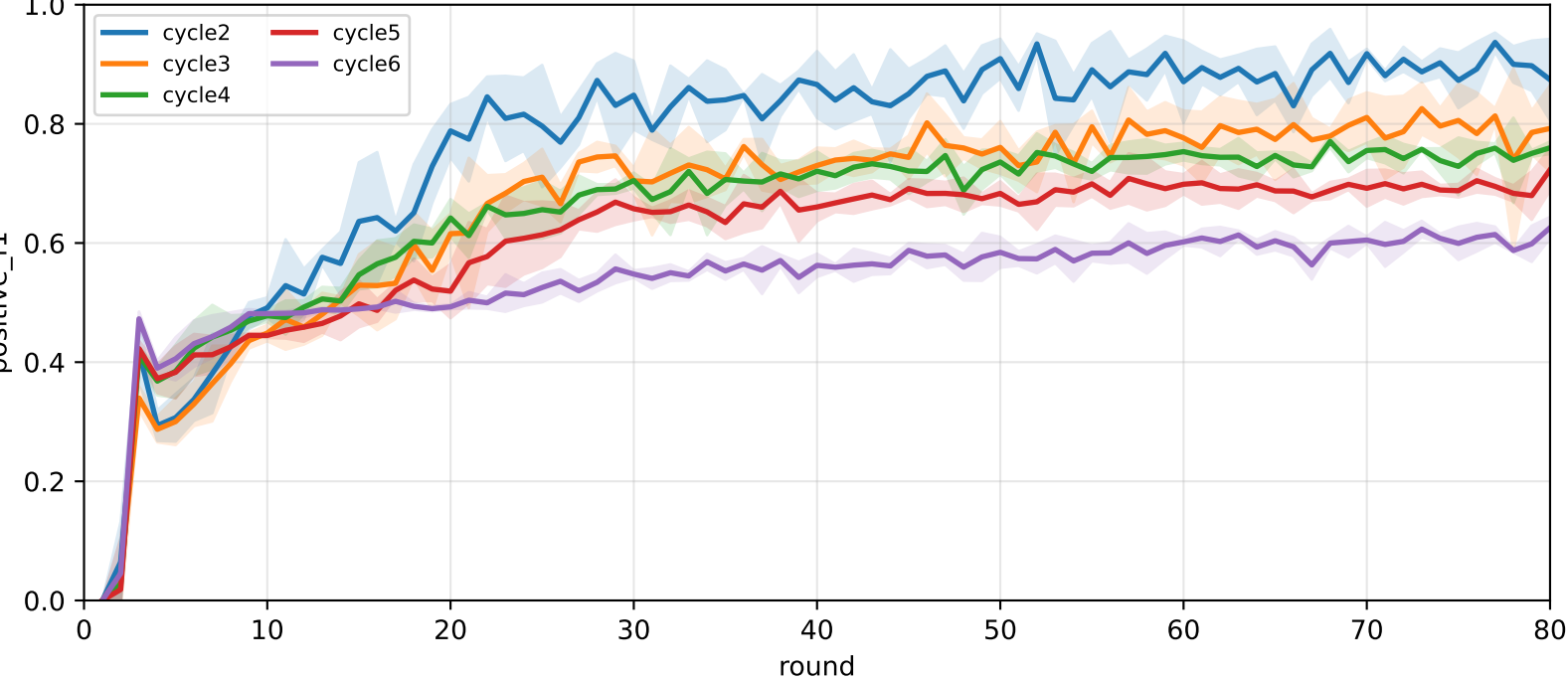
Pooled FedAvg positive F1



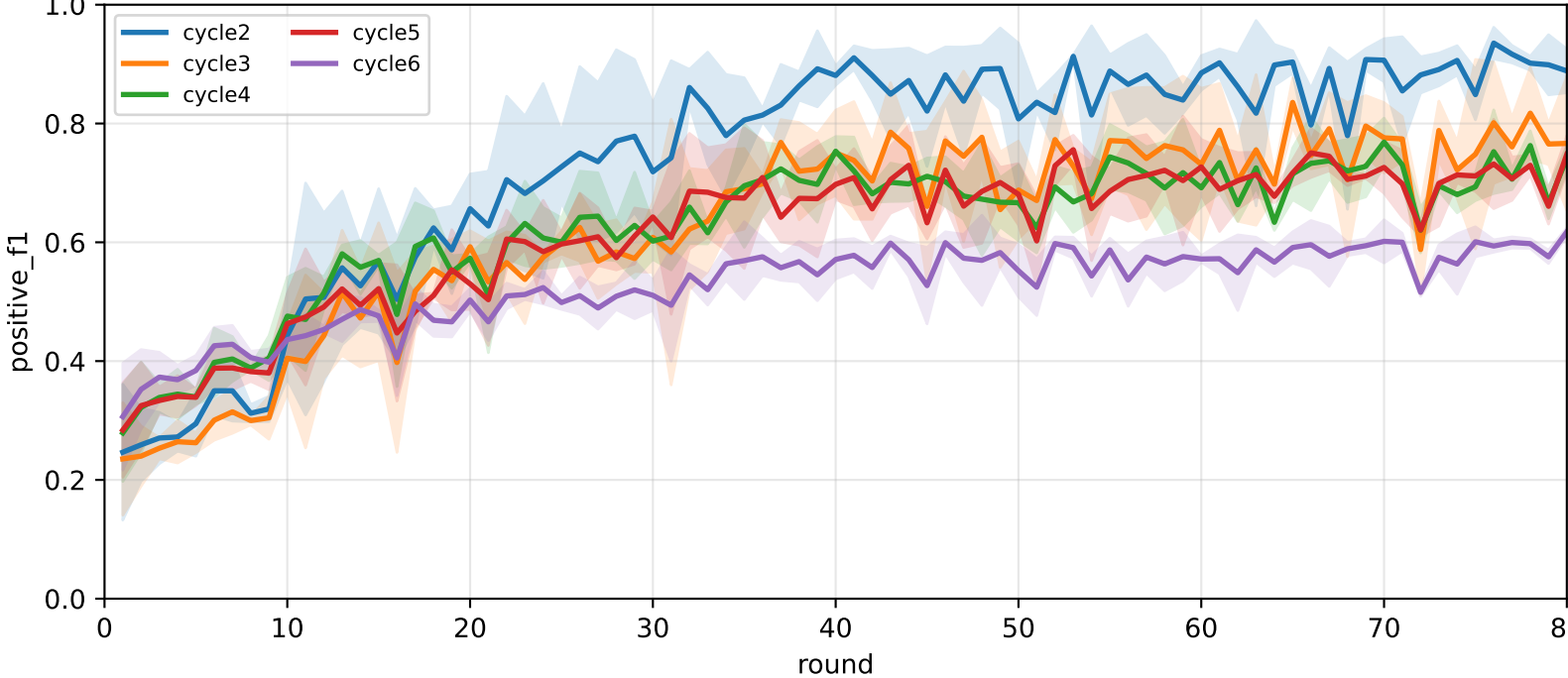
Pooled FedProx positive F1



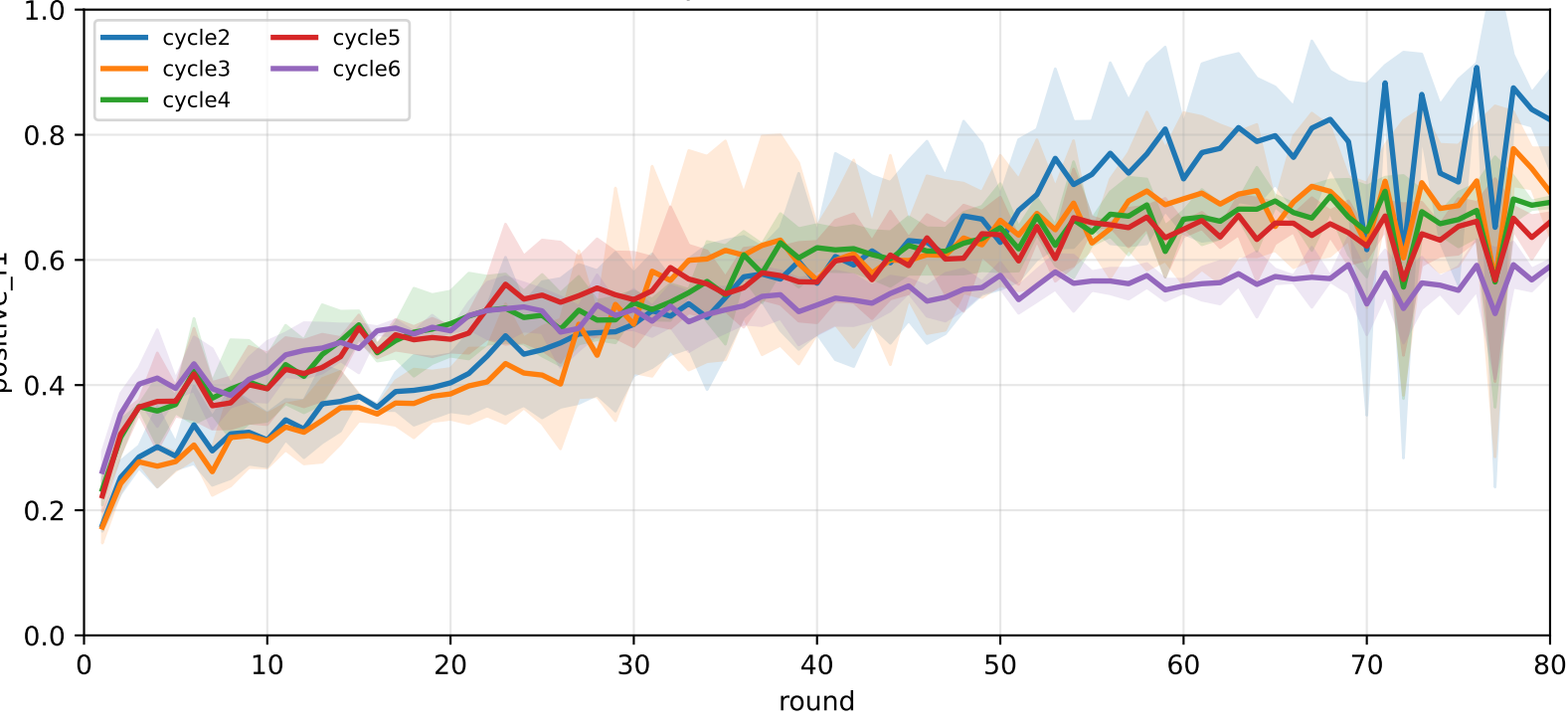
Pooled GCFL+ positive F1



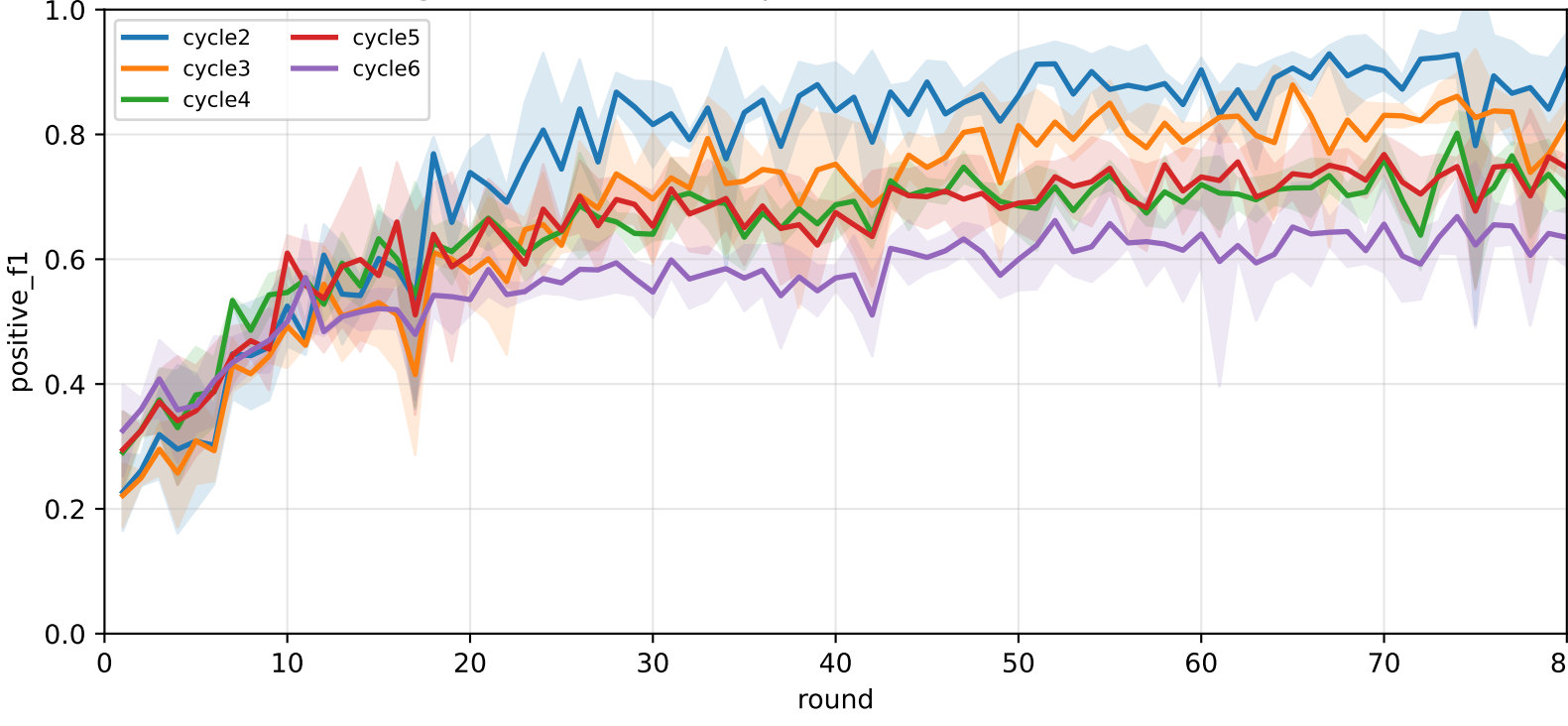
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



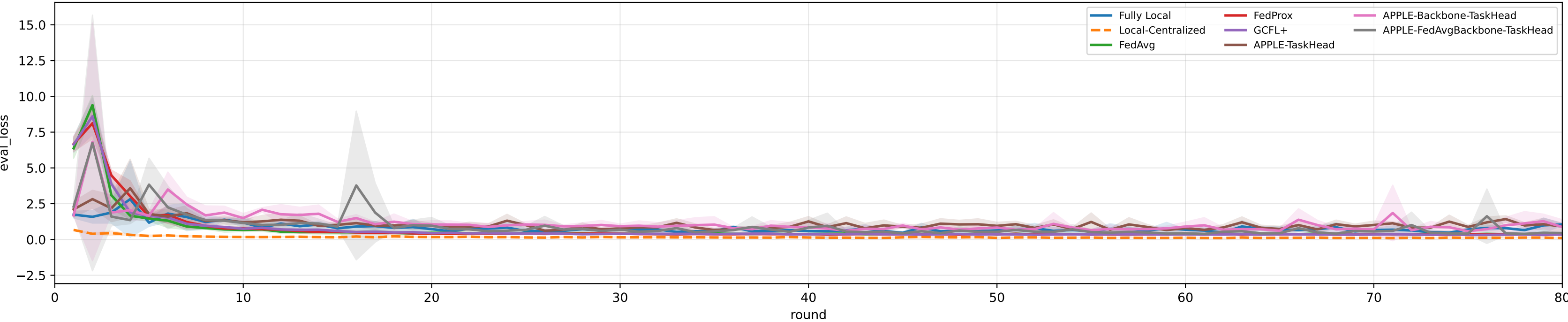
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q_task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

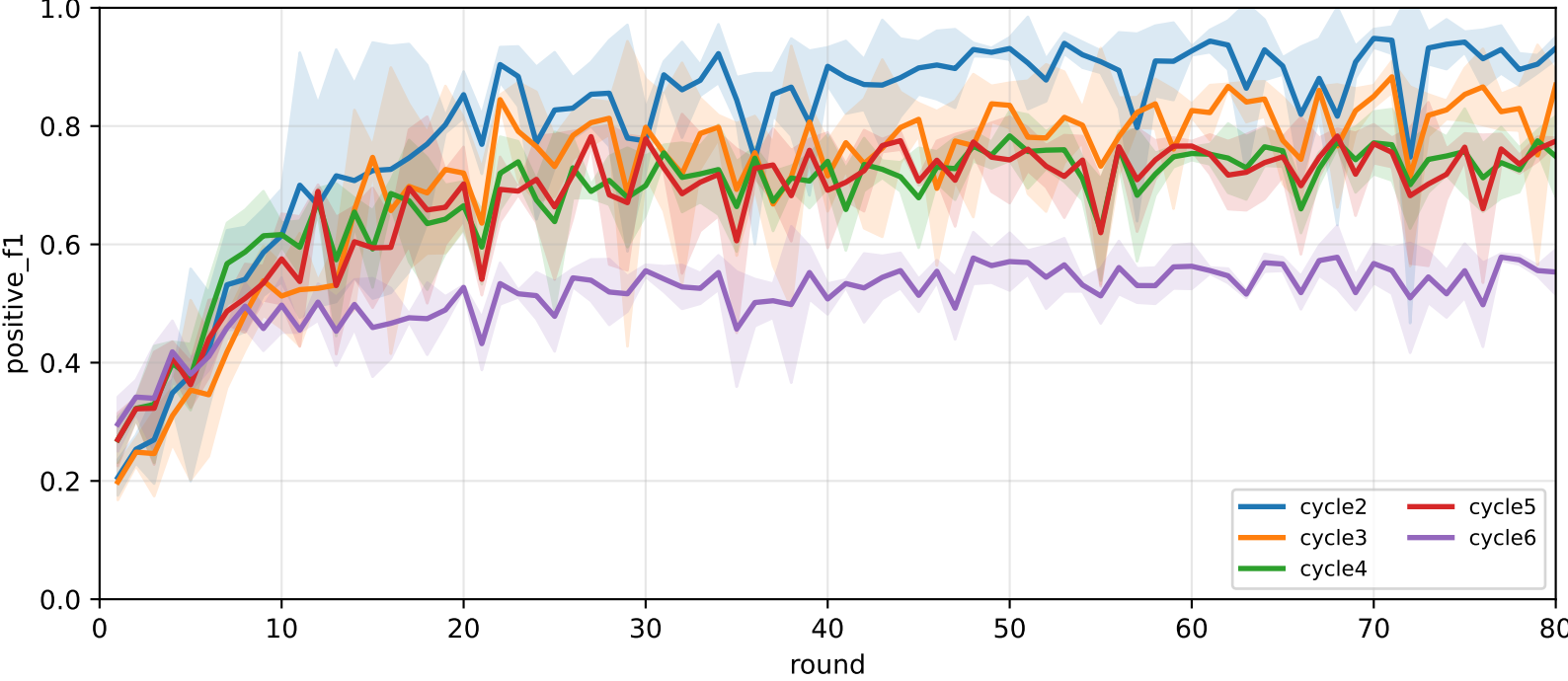
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.8% ± 3.3%	70.1% ± 2.7%	92.2% ± 3.4%	82.7% ± 3.5%	65.7% ± 8.3%	60.9% ± 7.0%	48.9% ± 2.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedAvg	69.8% ± 1.8%	73.1% ± 1.7%	91.5% ± 1.7%	82.7% ± 3.3%	71.2% ± 1.4%	63.3% ± 1.3%	57.0% ± 2.1%
FedProx	71.8% ± 1.1%	75.0% ± 1.4%	90.1% ± 3.9%	84.5% ± 2.5%	76.3% ± 1.8%	66.3% ± 0.7%	57.6% ± 0.3%
GCFL+	71.1% ± 1.1%	74.4% ± 0.8%	92.2% ± 2.0%	83.1% ± 2.6%	74.0% ± 1.0%	65.2% ± 1.3%	57.4% ± 1.1%
APPLE-TaskHead	69.5% ± 0.6%	71.5% ± 1.4%	82.4% ± 5.7%	79.0% ± 3.8%	71.0% ± 3.6%	67.5% ± 2.3%	57.7% ± 1.0%
APPLE-Backbone-TaskHead	62.7% ± 3.8%	63.7% ± 4.1%	65.2% ± 12.0%	69.9% ± 0.9%	67.1% ± 3.2%	61.4% ± 3.5%	54.9% ± 3.5%
APPLE-FedAvgBackbone-TaskHead	73.2% ± 2.1%	76.1% ± 2.0%	91.3% ± 2.0%	86.8% ± 2.1%	71.3% ± 5.3%	72.3% ± 0.7%	59.0% ± 1.1%

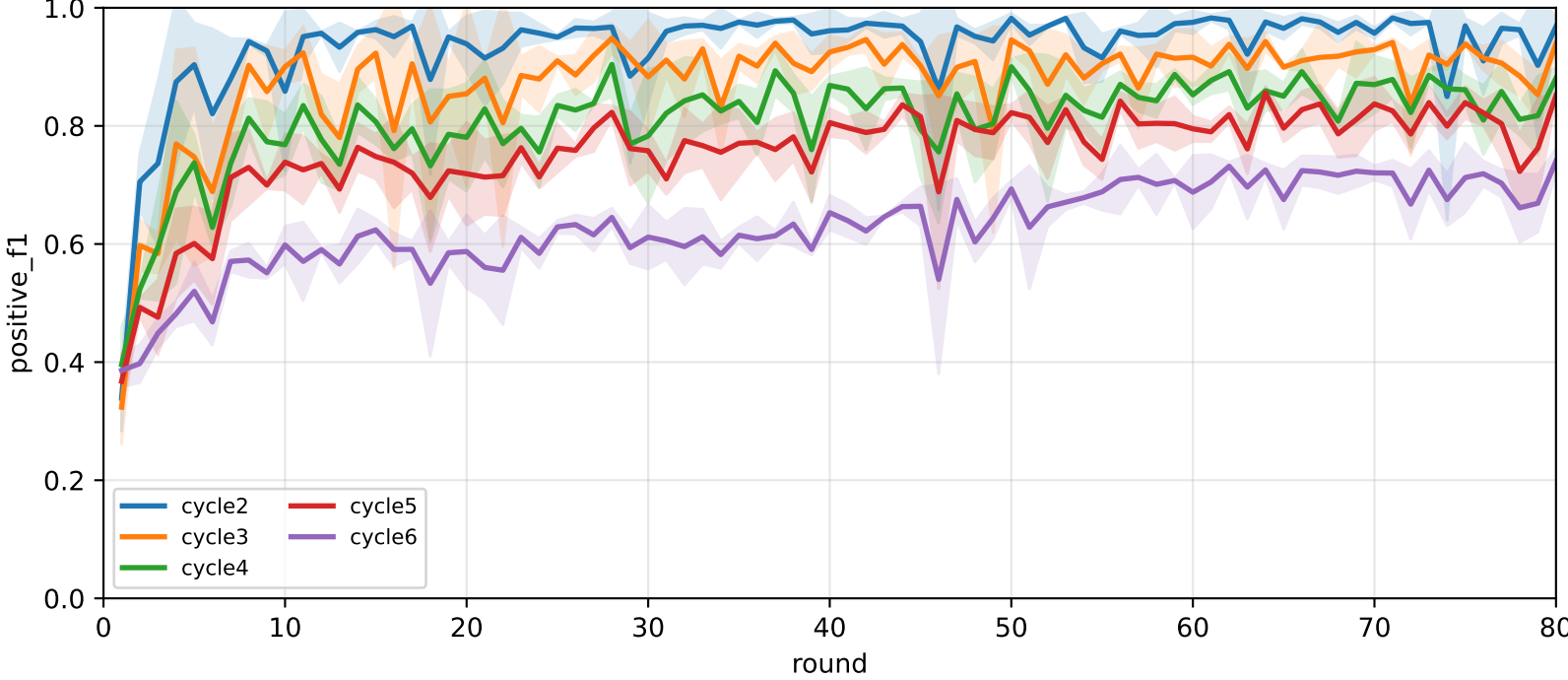
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



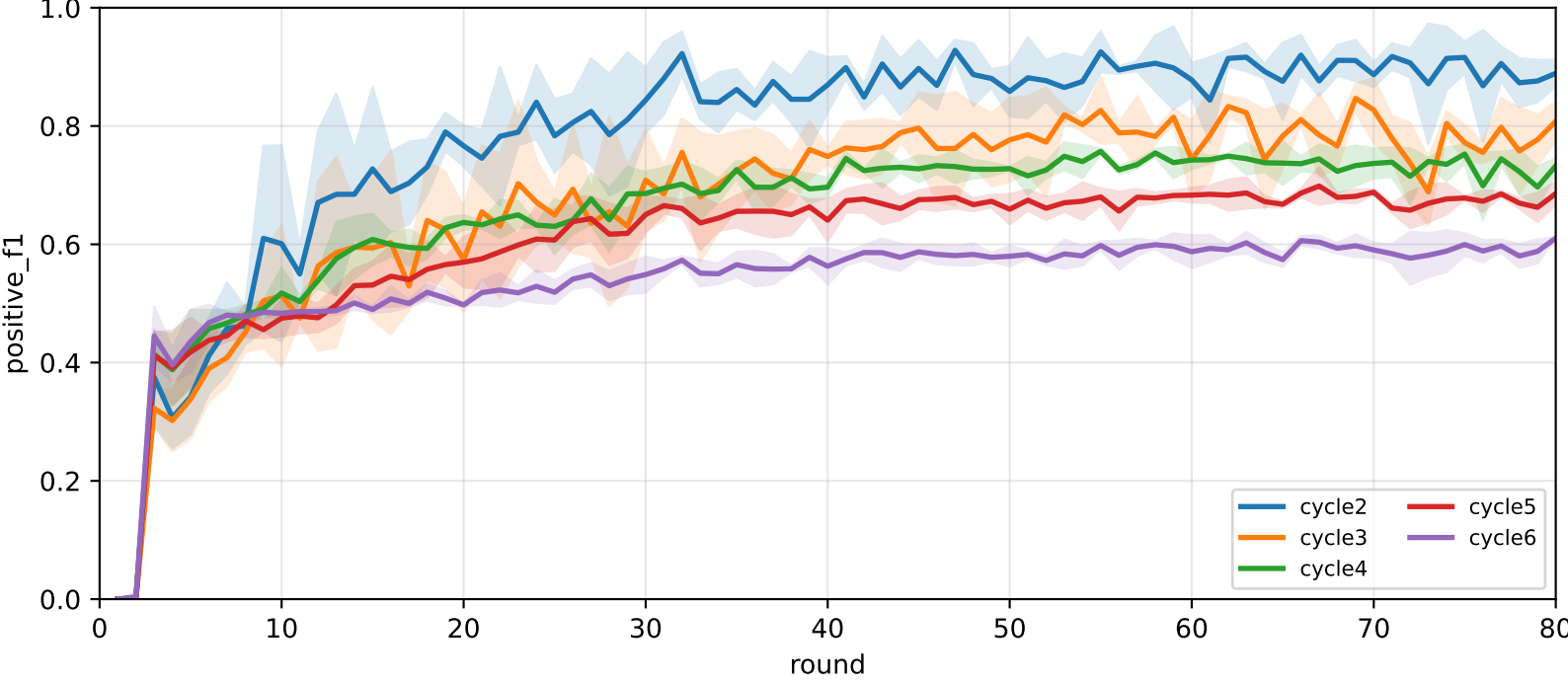
Pooled Fully Local positive F1



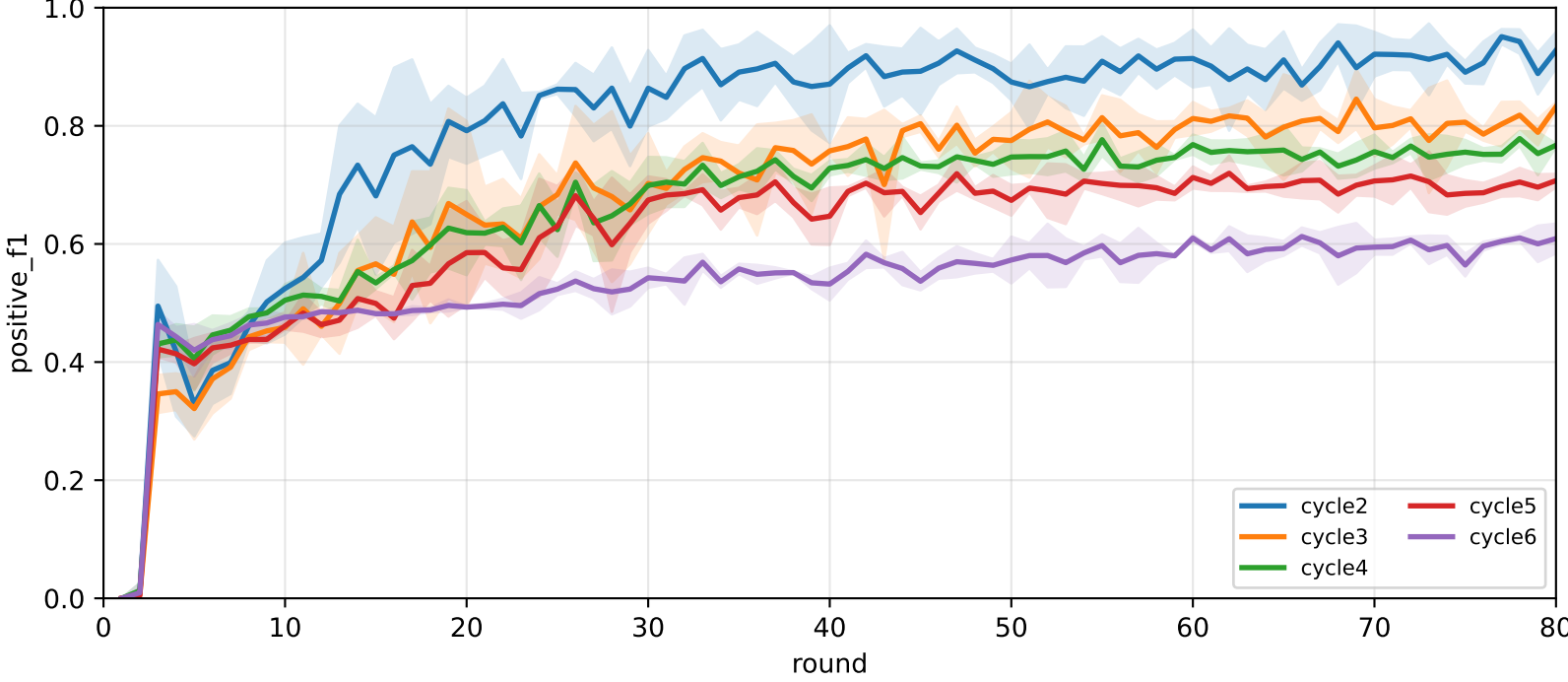
Pooled Local-Centralized positive F1



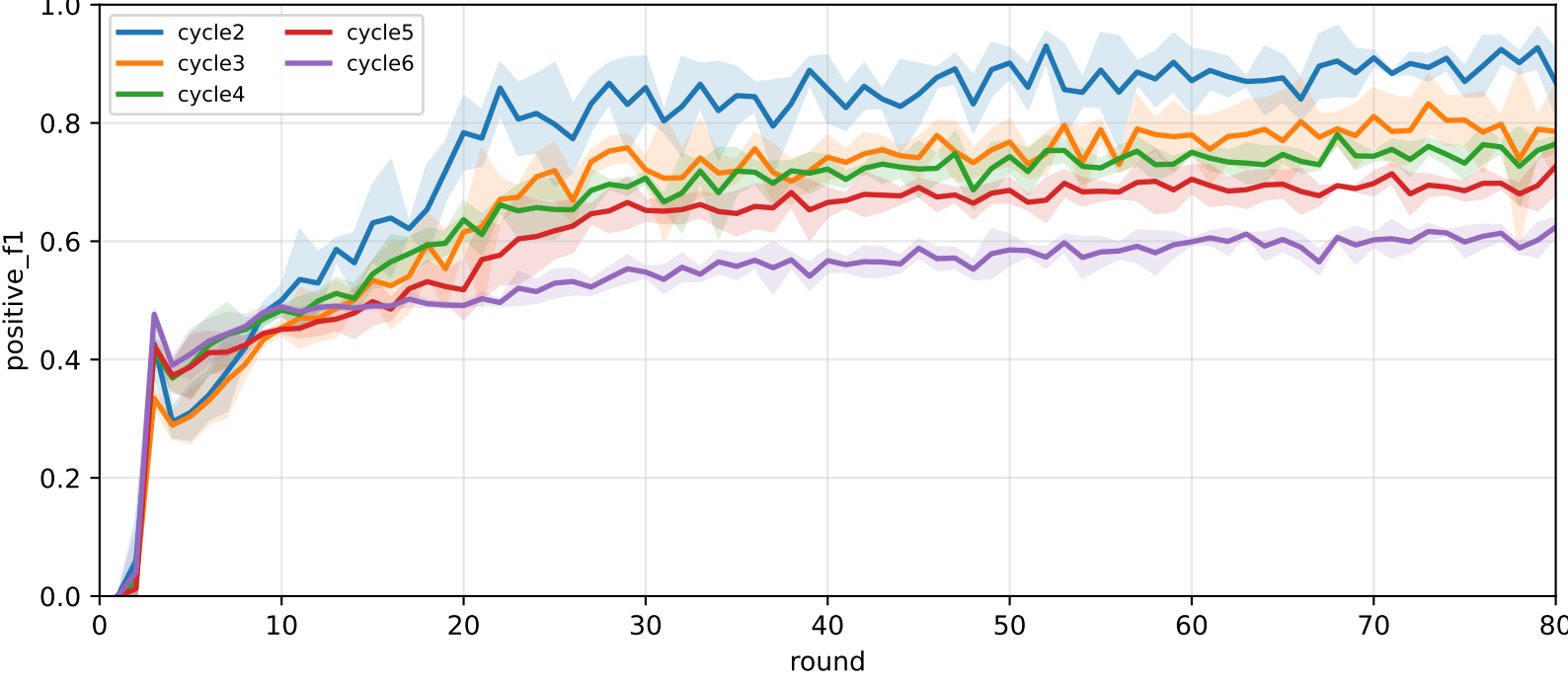
Pooled FedAvg positive F1



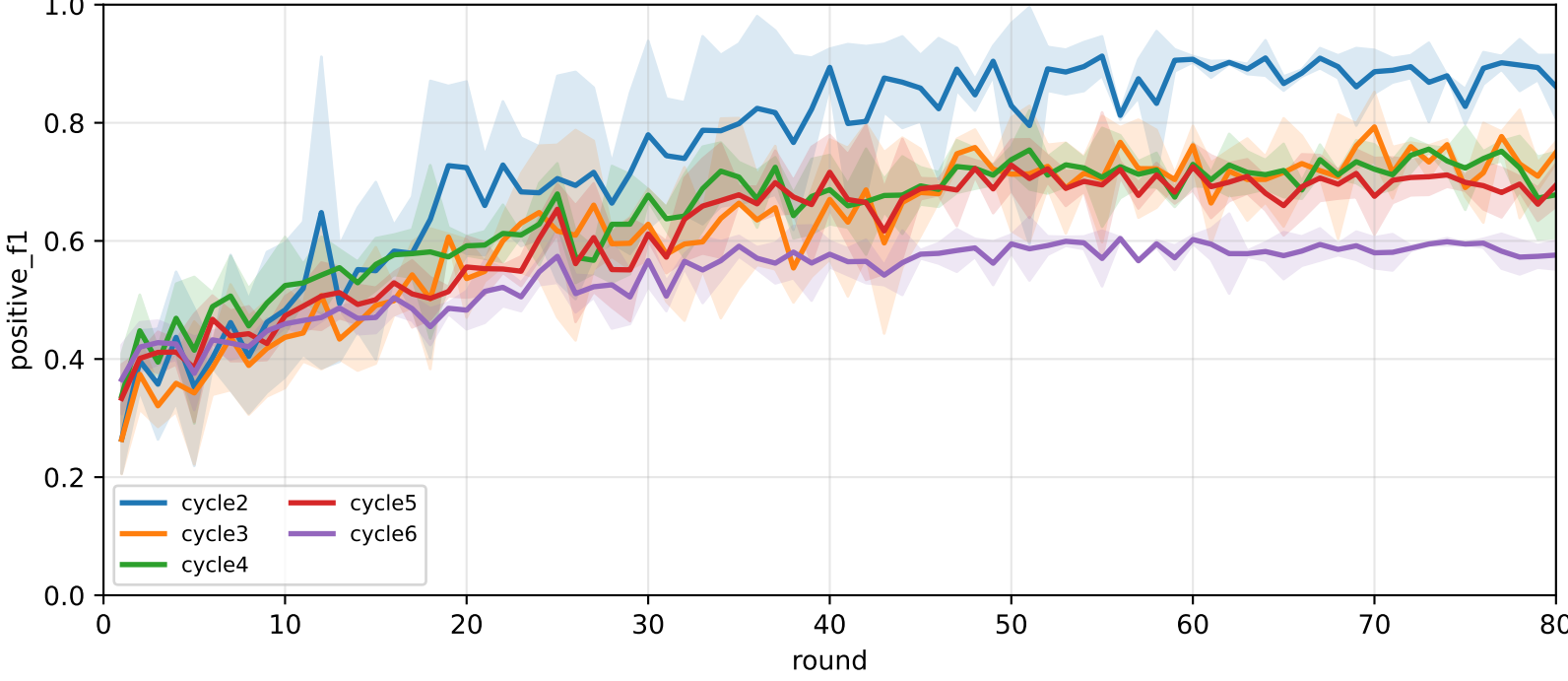
Pooled FedProx positive F1



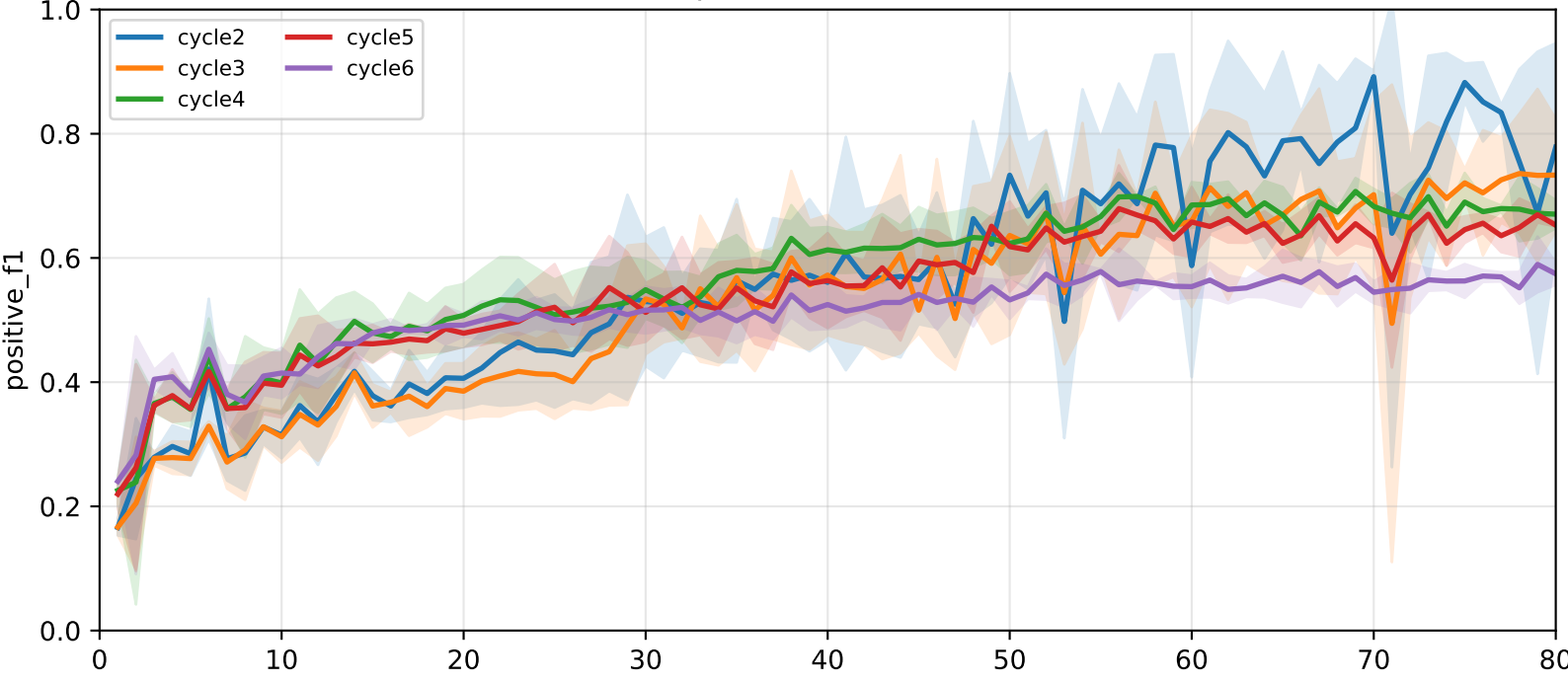
Pooled GCFL+ positive F1



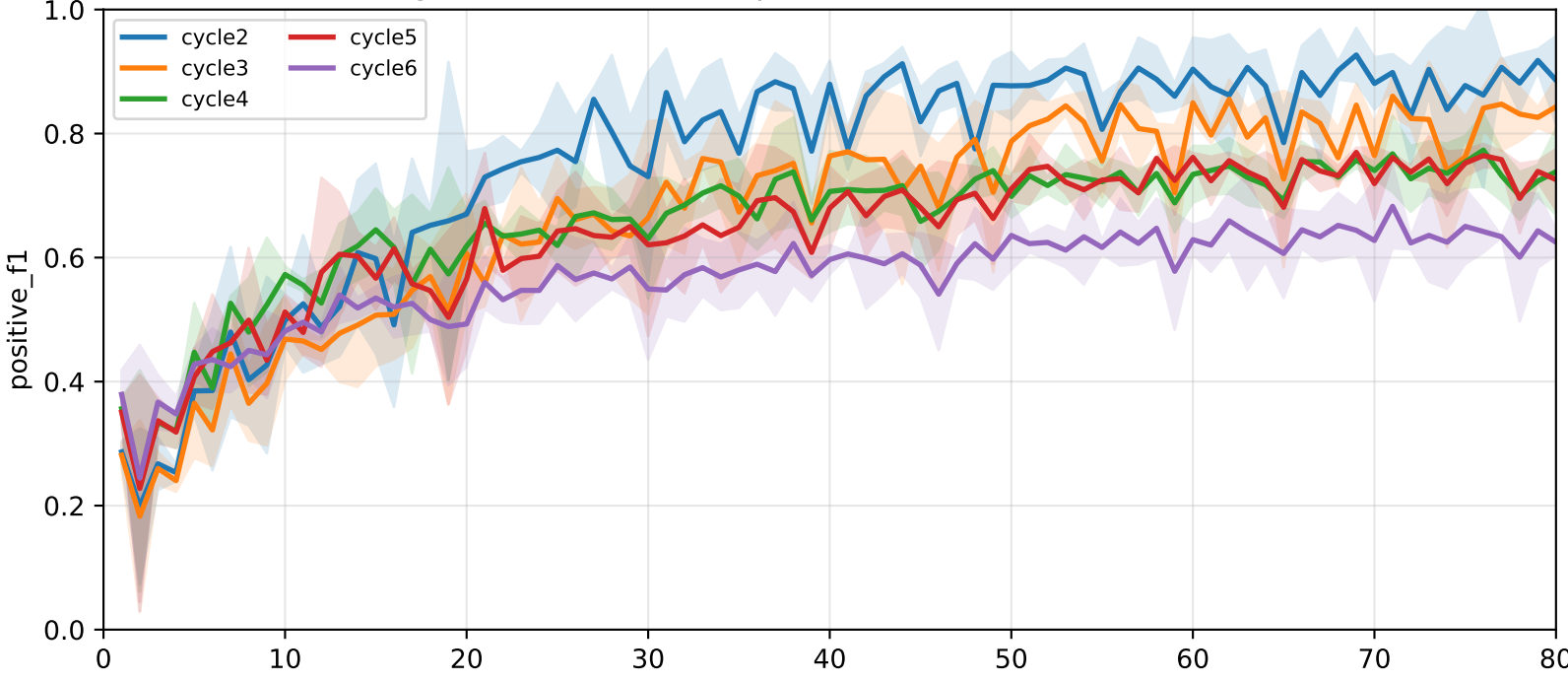
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



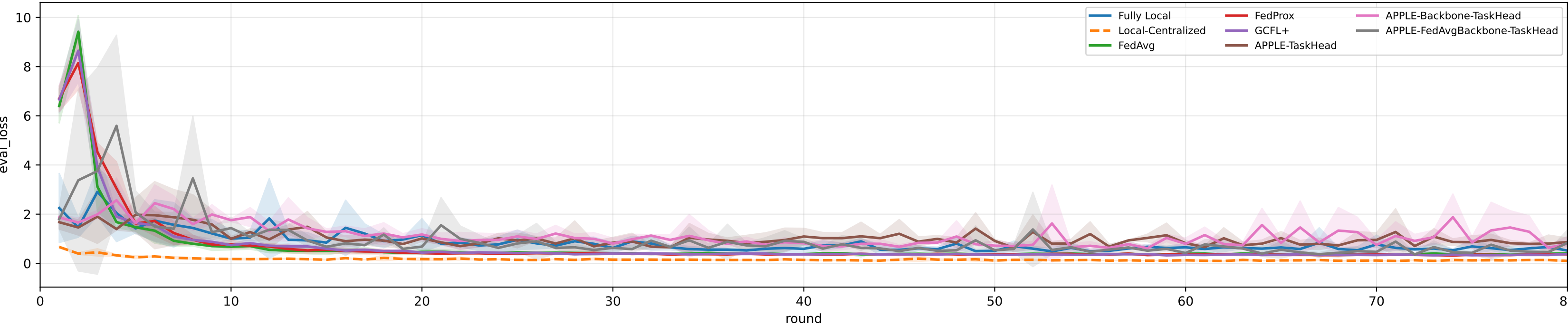
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q_task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

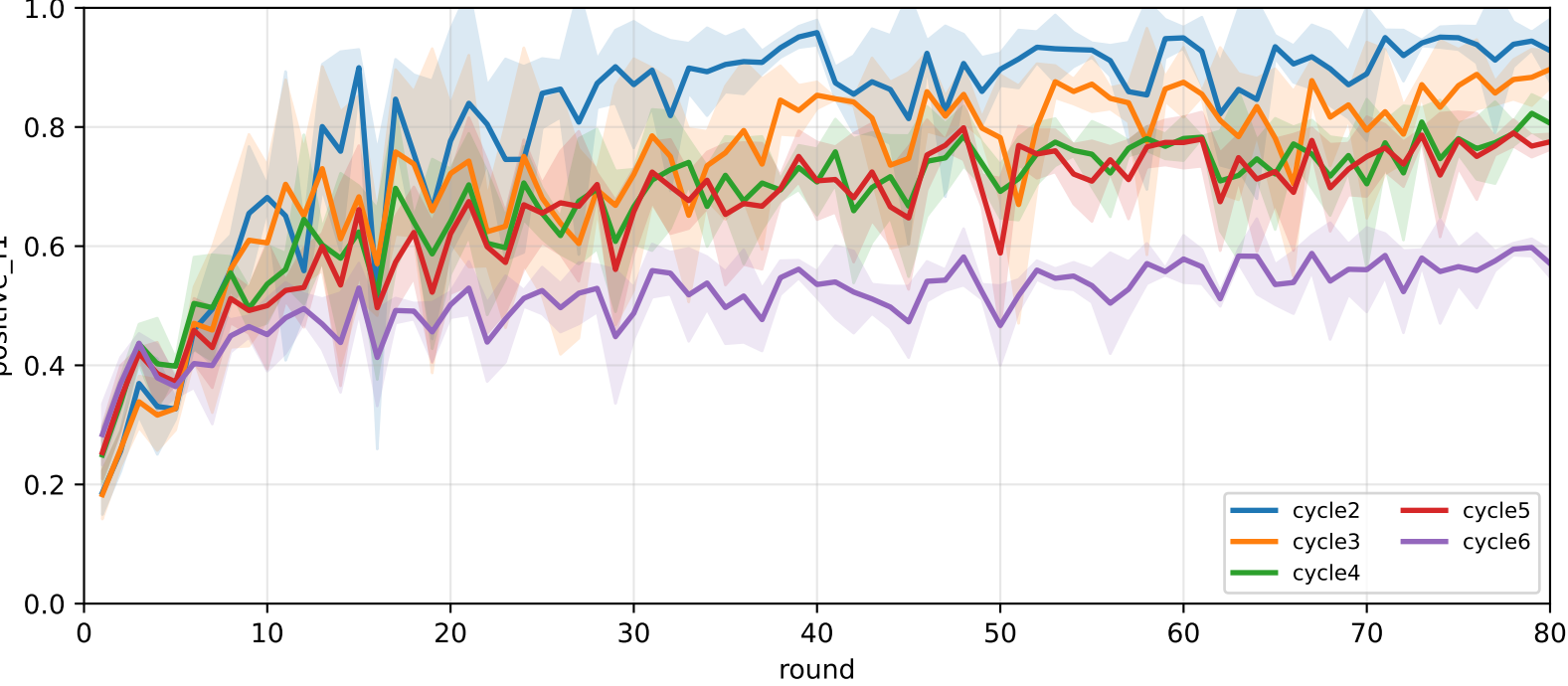
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	70.5% ± 1.9%	74.2% ± 2.2%	87.8% ± 6.8%	<u>86.4% ± 2.4%</u>	74.0% ± 2.0%	69.7% ± 1.8%	53.3% ± 1.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedAvg	70.3% ± 1.1%	73.4% ± 0.8%	89.6% ± 2.1%	83.4% ± 2.9%	72.9% ± 1.2%	64.1% ± 0.9%	57.2% ± 1.6%
FedProx	<u>72.2% ± 0.8%</u>	<u>75.3% ± 1.3%</u>	89.9% ± 4.4%	85.3% ± 2.4%	<u>76.5% ± 1.6%</u>	66.8% ± 0.9%	58.1% ± 0.6%
GCFL+	70.9% ± 0.9%	74.3% ± 0.5%	<u>92.4% ± 2.9%</u>	82.7% ± 2.8%	75.0% ± 0.6%	64.5% ± 1.1%	56.9% ± 0.9%
APPLE-TaskHead	65.9% ± 7.2%	68.5% ± 6.6%	87.1% ± 1.9%	70.1% ± 9.7%	66.8% ± 7.8%	63.8% ± 9.5%	54.8% ± 5.6%
APPLE-Backbone-TaskHead	62.1% ± 3.7%	63.4% ± 4.0%	69.4% ± 6.6%	66.8% ± 1.0%	66.4% ± 4.9%	61.2% ± 5.5%	53.0% ± 1.9%
APPLE-FedAvgBackbone-TaskHead	71.1% ± 1.5%	73.6% ± 1.5%	90.2% ± 2.6%	74.9% ± 5.1%	73.0% ± 3.0%	<u>71.6% ± 0.6%</u>	<u>58.6% ± 1.2%</u>

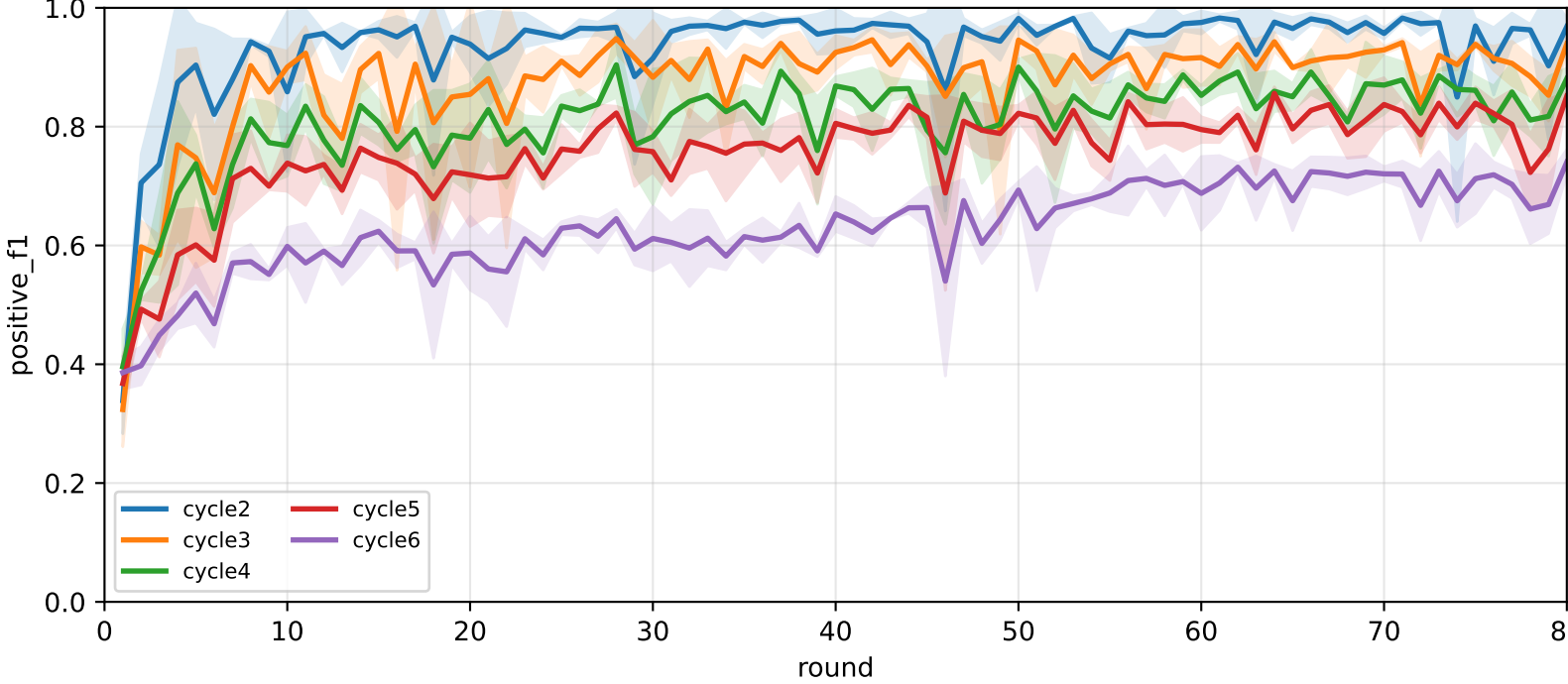
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



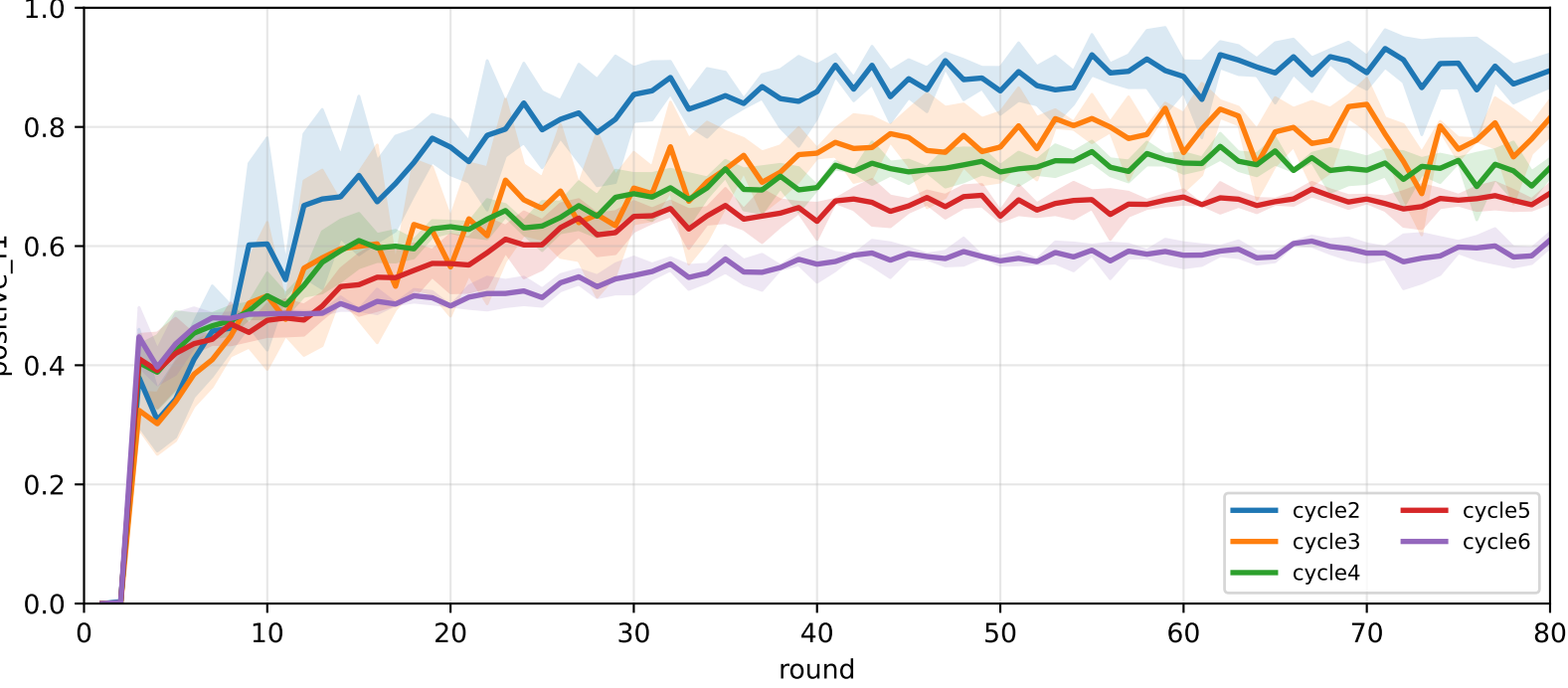
Pooled Fully Local positive F1



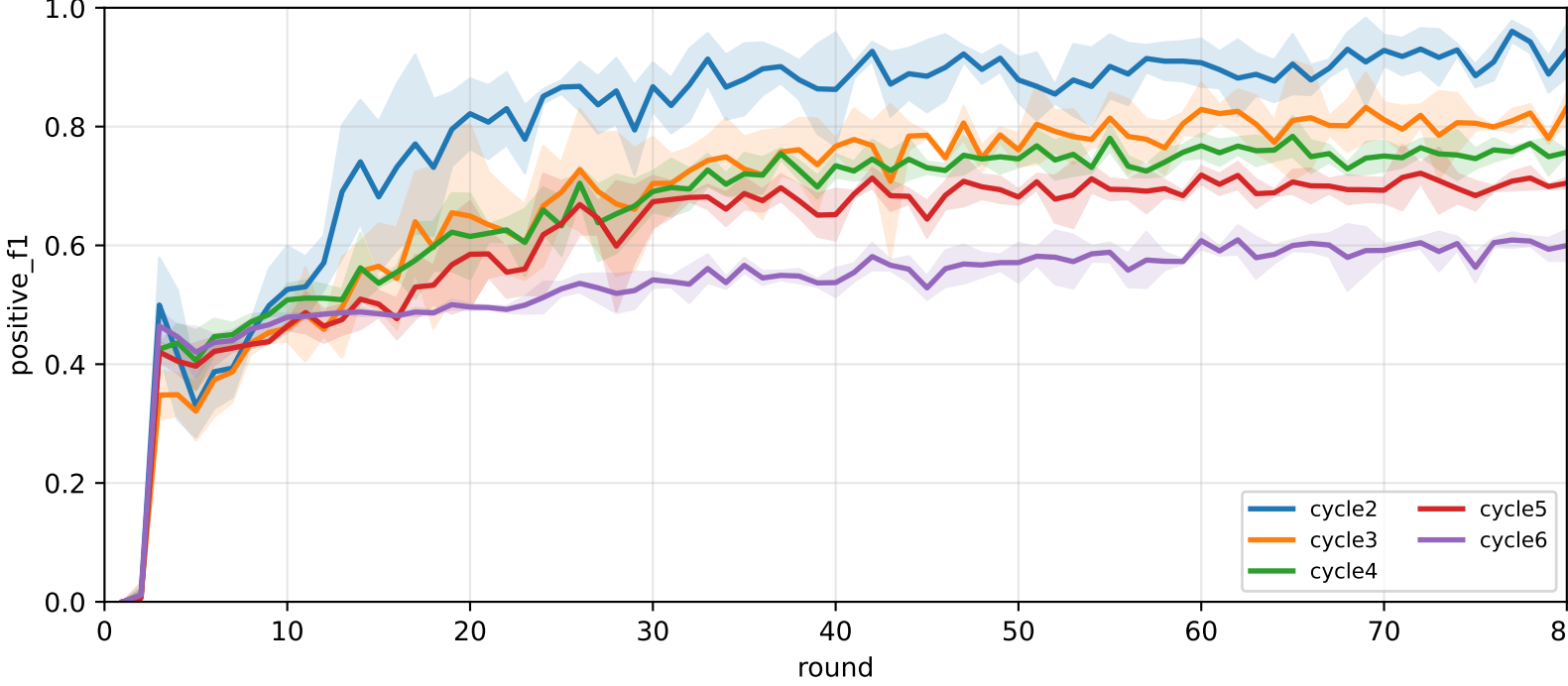
Pooled Local-Centralized positive F1



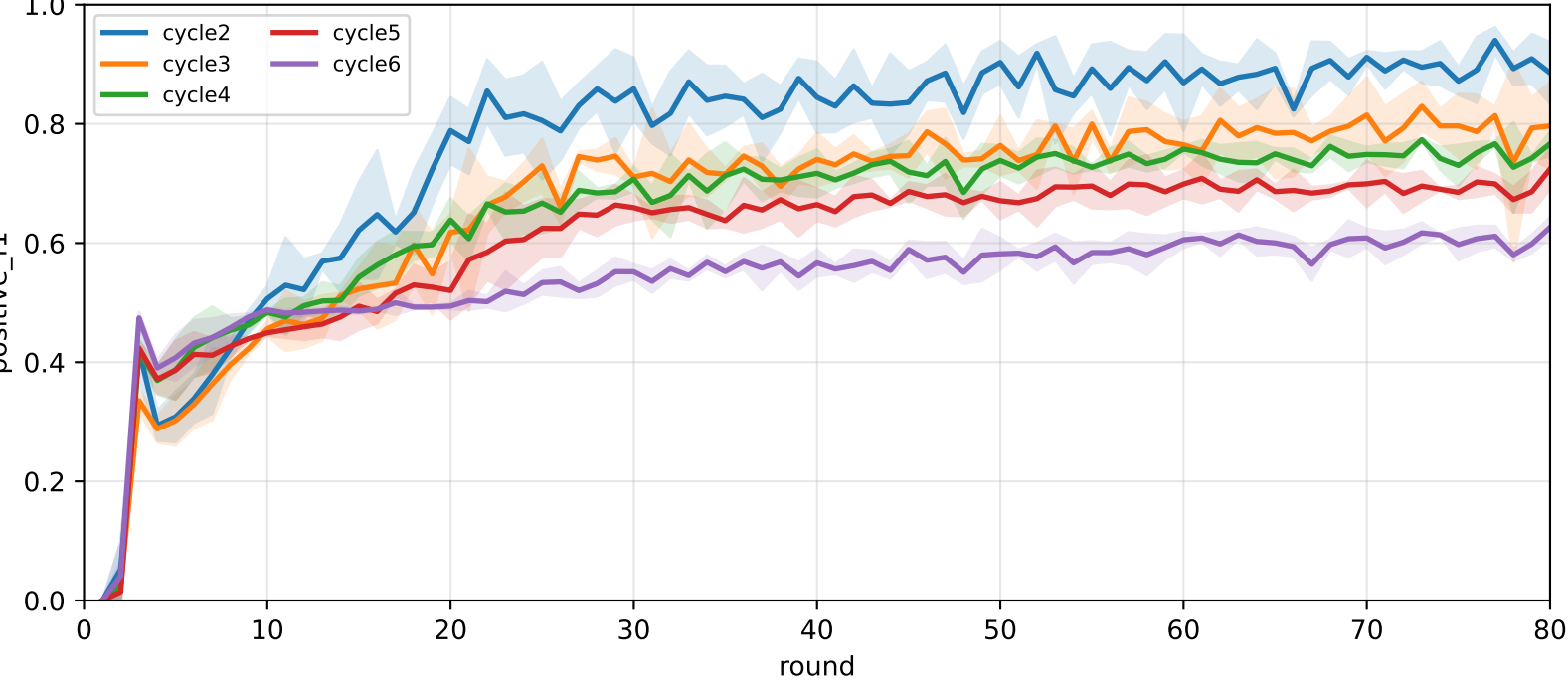
Pooled FedAvg positive F1



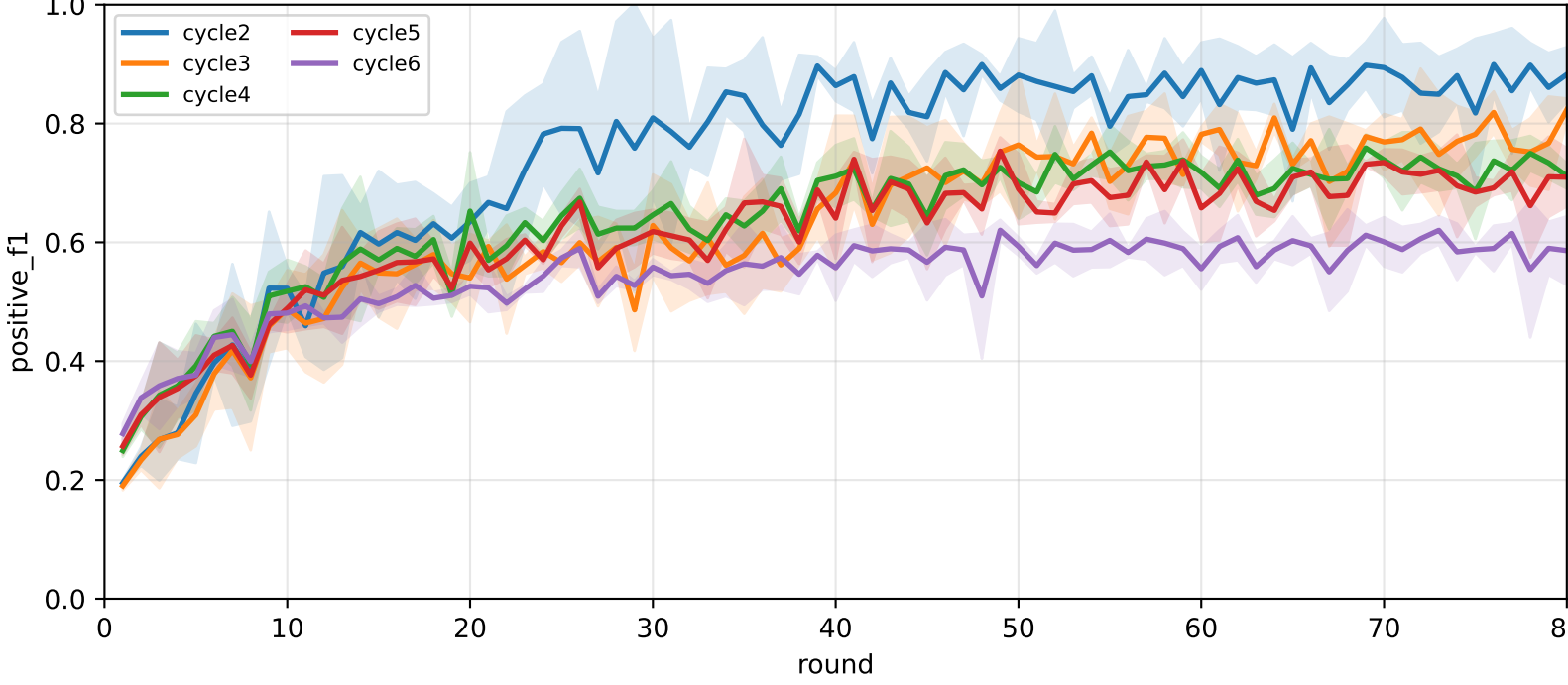
Pooled FedProx positive F1



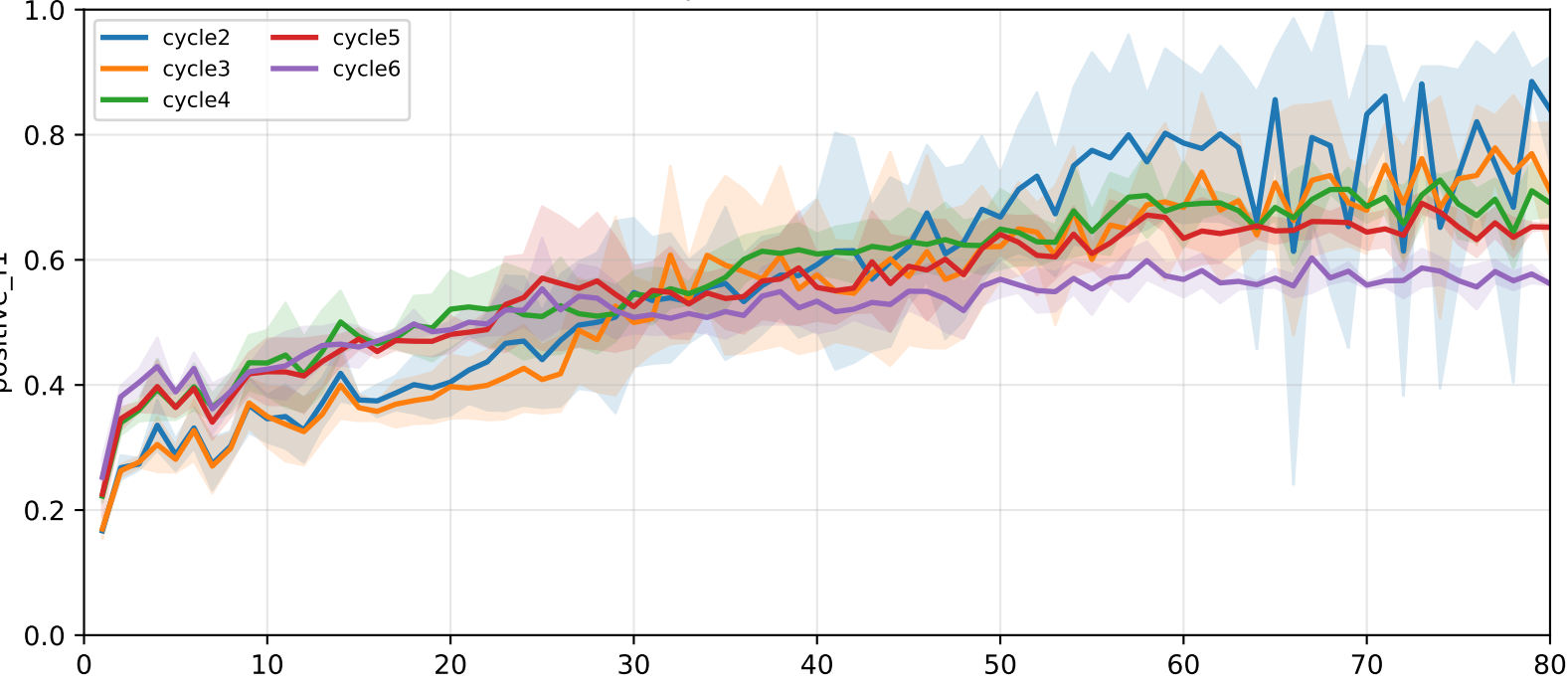
Pooled GCFL+ positive F1



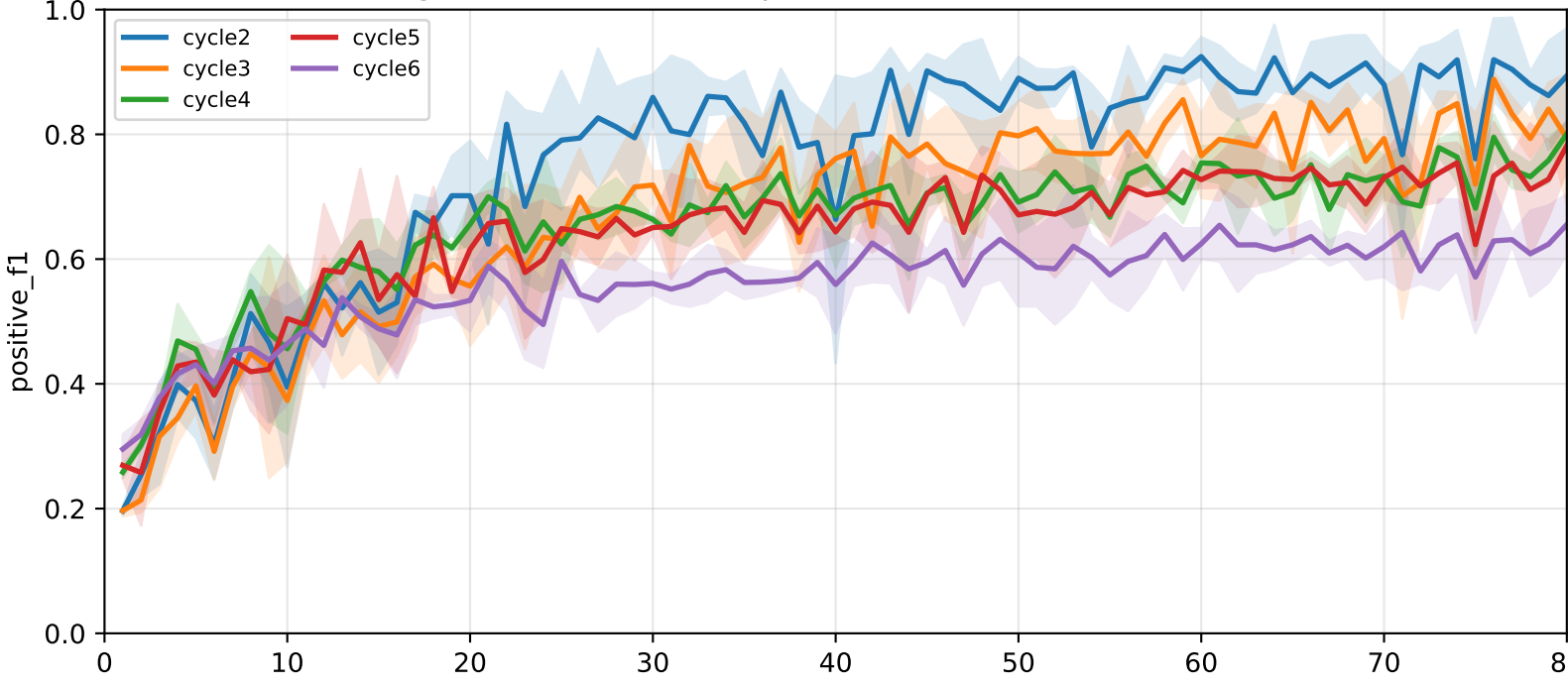
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



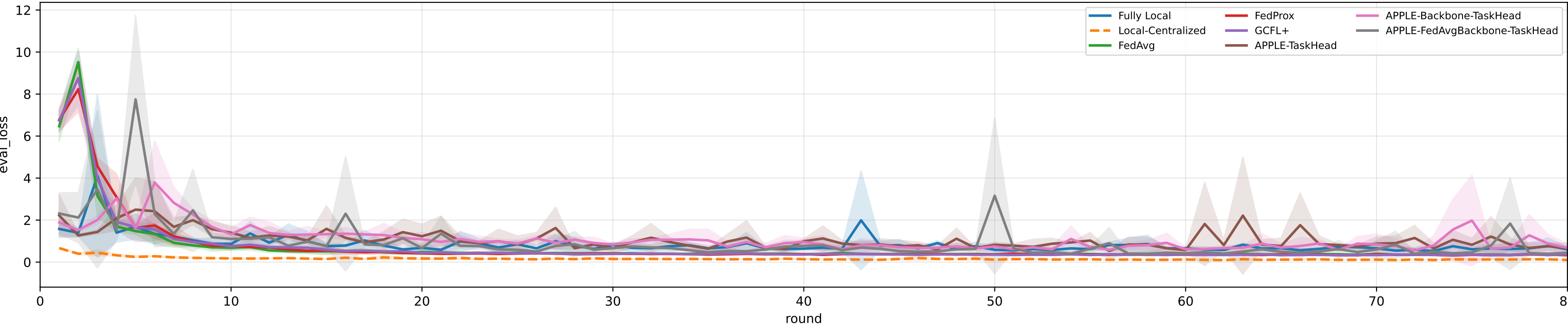
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

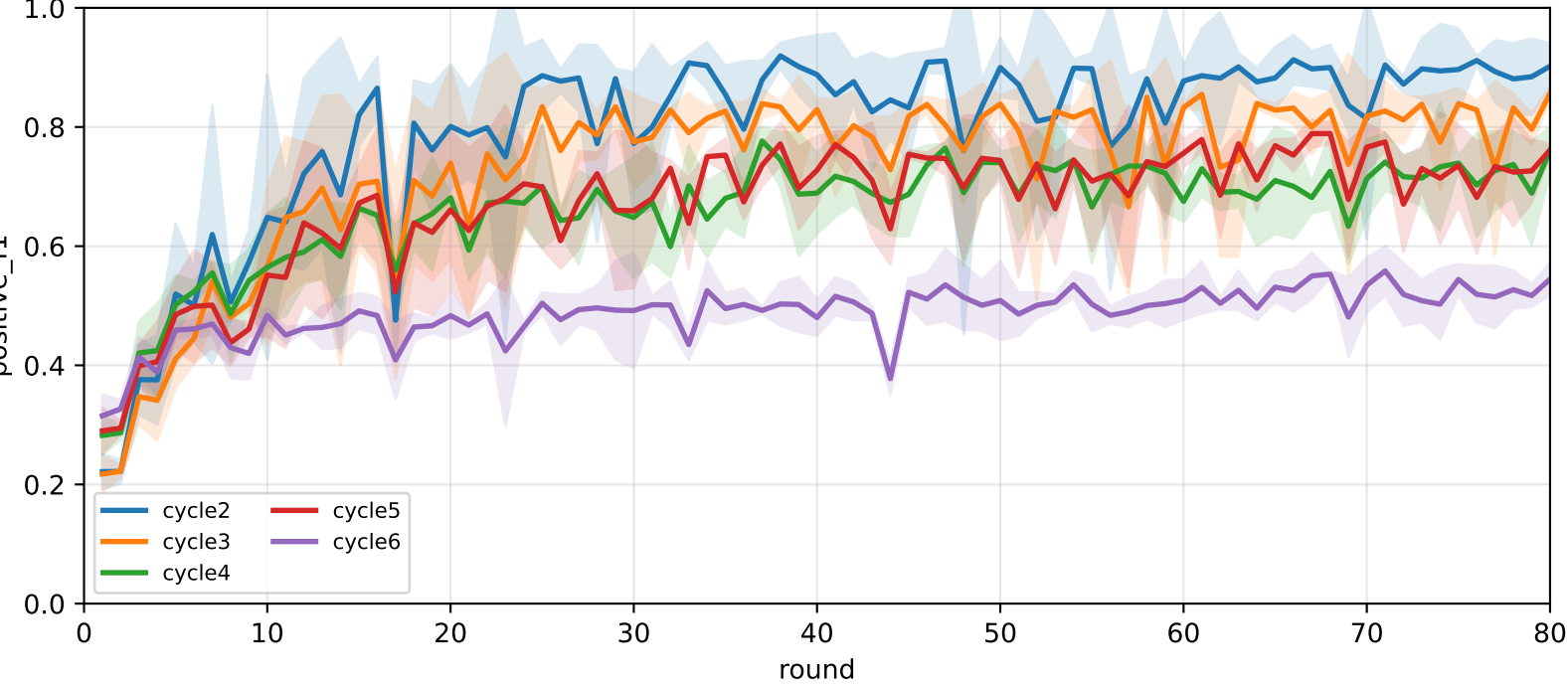
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.5% ± 3.5%	69.6% ± 3.3%	87.3% ± 2.7%	84.4% ± 3.4%	62.5% ± 7.2%	67.0% ± 5.0%	46.8% ± 1.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
FedAvg	70.1% ± 1.0%	73.5% ± 1.1%	91.7% ± 2.3%	83.0% ± 3.3%	72.2% ± 0.9%	64.2% ± 0.5%	56.6% ± 1.6%
FedProx	71.9% ± 1.7%	<u>75.0% ± 1.9%</u>	89.8% ± 3.5%	<u>85.0% ± 1.6%</u>	<u>76.2% ± 2.7%</u>	66.3% ± 1.7%	57.8% ± 1.0%
GCFL+	71.4% ± 1.4%	74.8% ± 0.9%	<u>92.9% ± 3.1%</u>	83.0% ± 3.1%	75.8% ± 1.4%	65.1% ± 1.6%	57.2% ± 1.6%
APPLE-TaskHead	71.0% ± 2.8%	73.4% ± 2.6%	87.4% ± 5.8%	80.2% ± 5.6%	70.0% ± 3.2%	71.8% ± 2.0%	57.7% ± 2.3%
APPLE-Backbone-TaskHead	65.3% ± 3.9%	67.2% ± 4.6%	76.0% ± 13.9%	73.0% ± 4.0%	68.0% ± 3.9%	63.8% ± 3.9%	55.2% ± 2.4%
APPLE-FedAvgBackbone-TaskHead	<u>72.7% ± 2.0%</u>	74.4% ± 1.6%	82.6% ± 3.0%	83.3% ± 2.1%	72.7% ± 2.6%	<u>72.1% ± 2.8%</u>	<u>61.4% ± 3.0%</u>

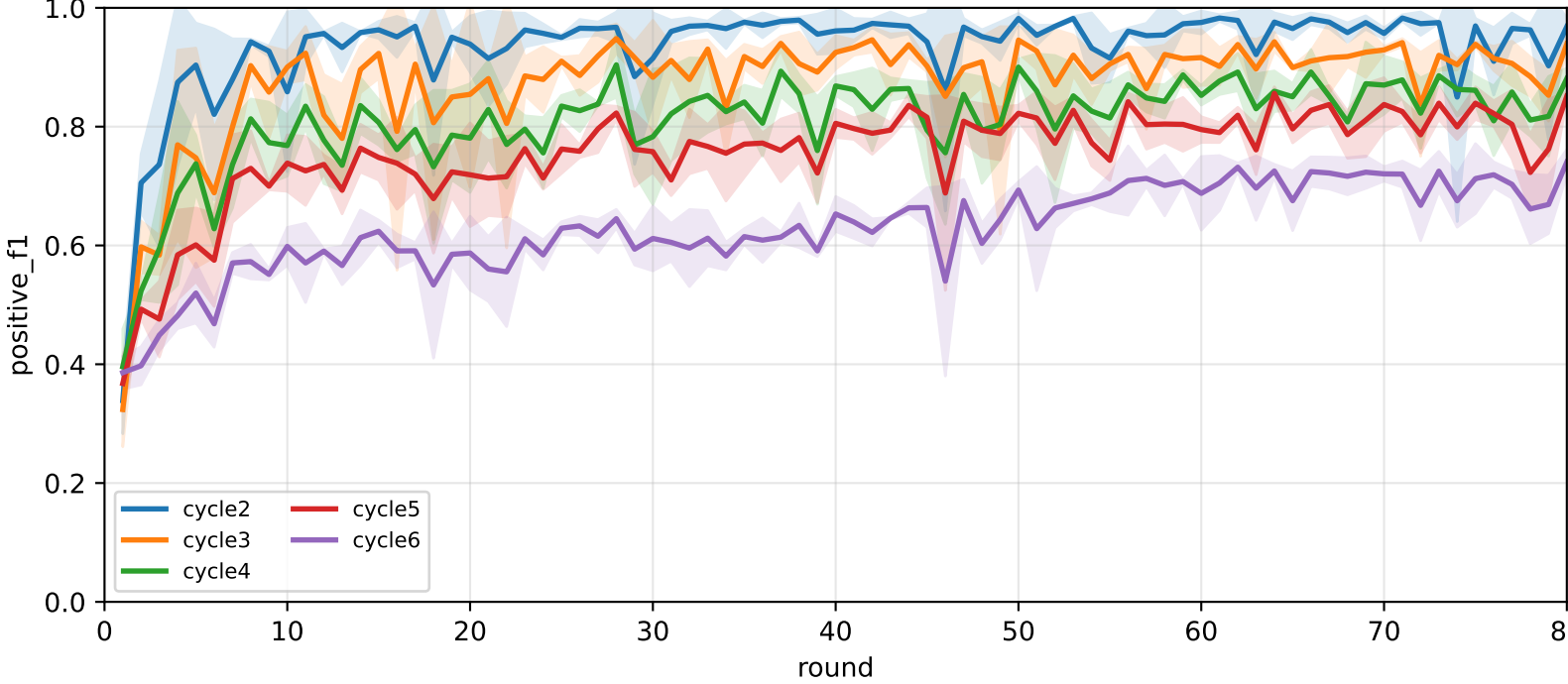
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



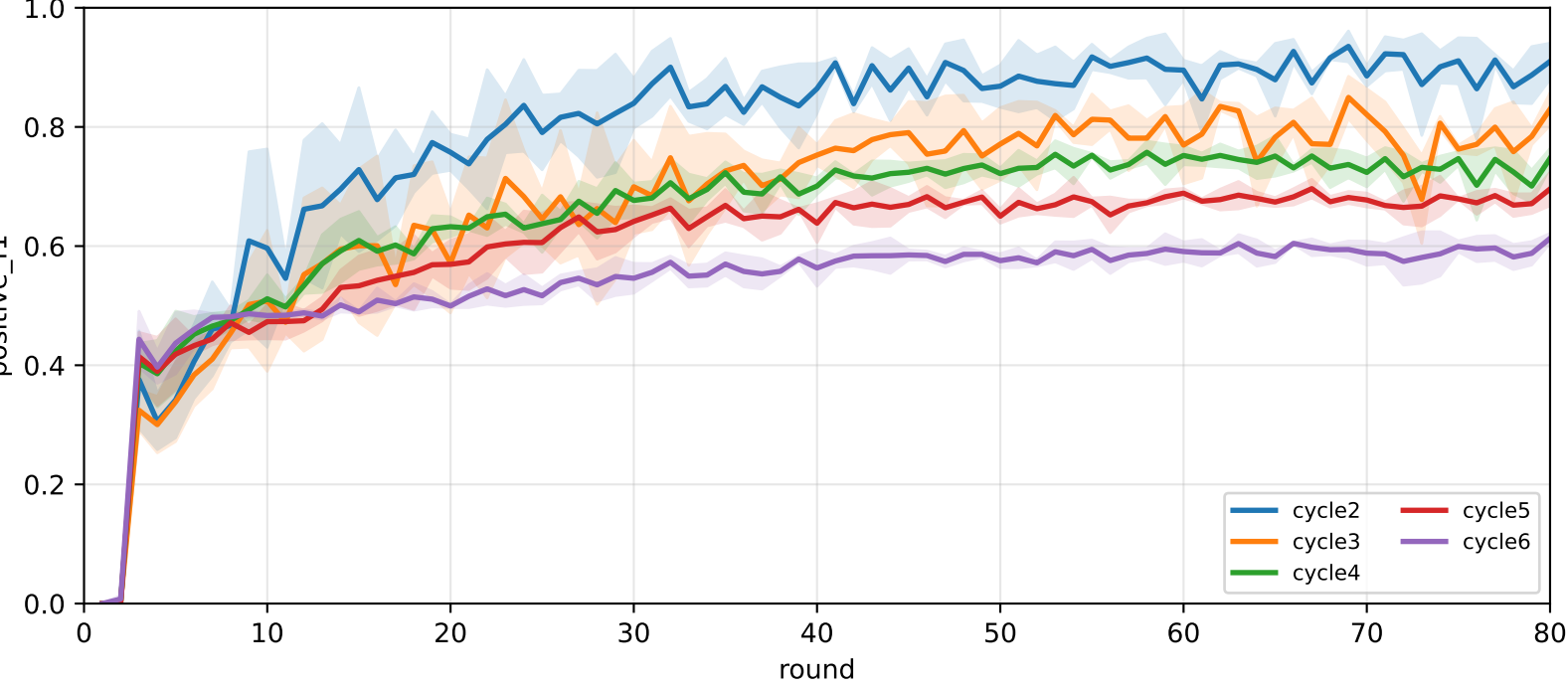
Pooled Fully Local positive F1



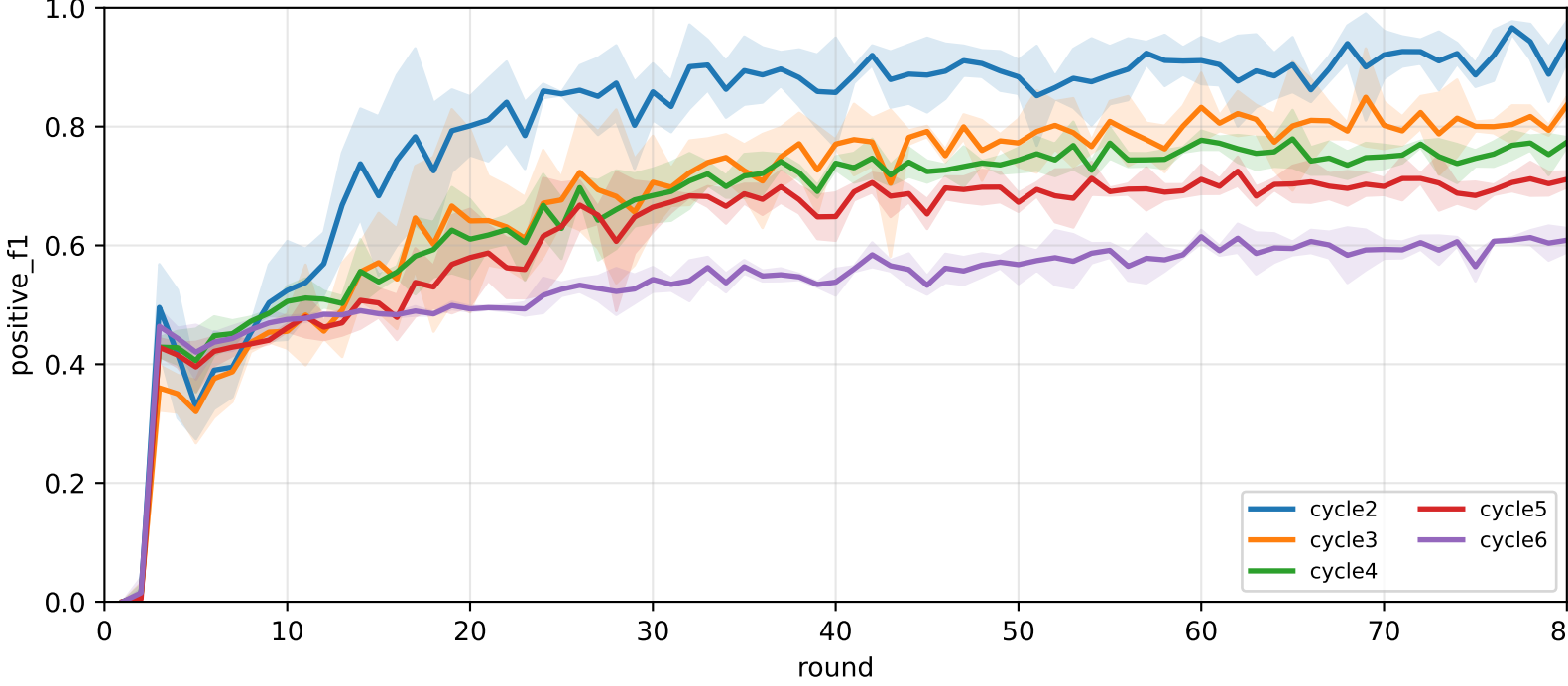
Pooled Local-Centralized positive F1



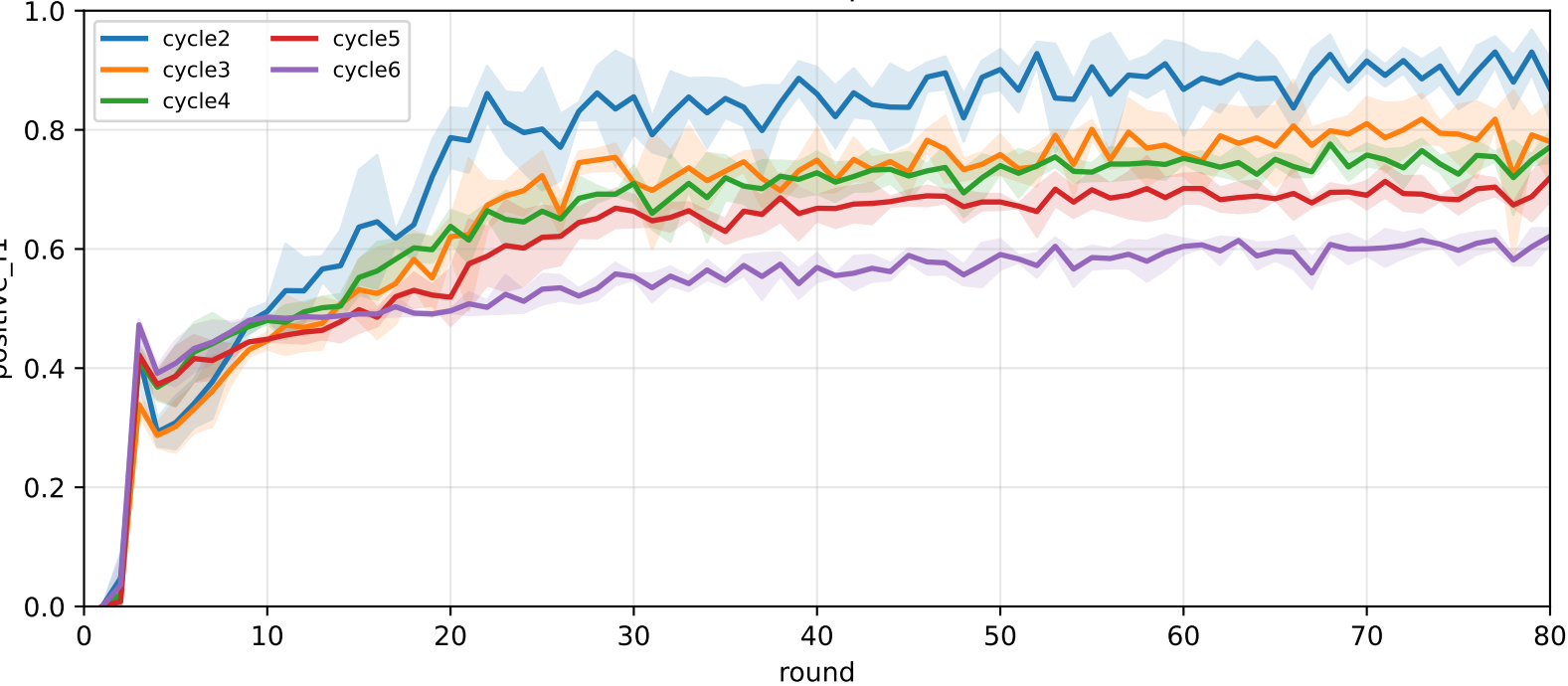
Pooled FedAvg positive F1



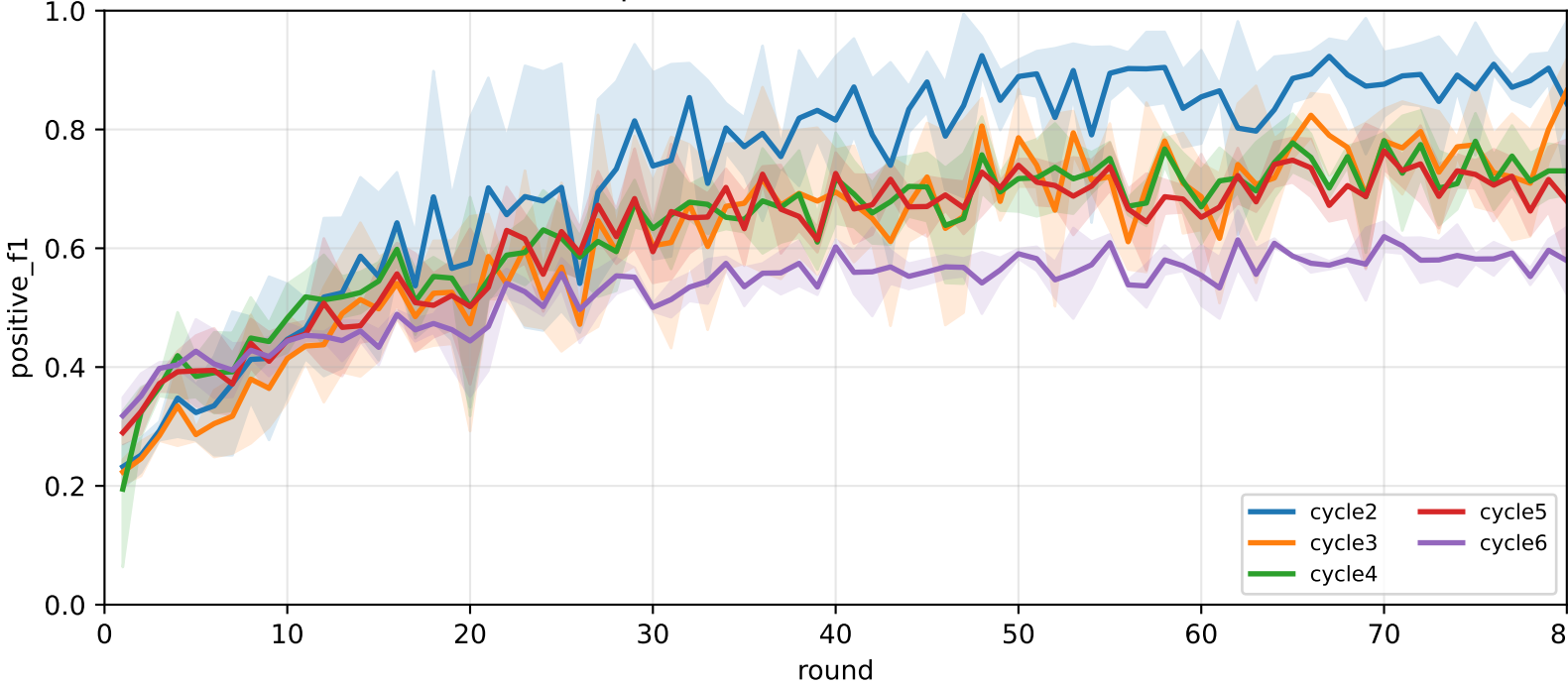
Pooled FedProx positive F1



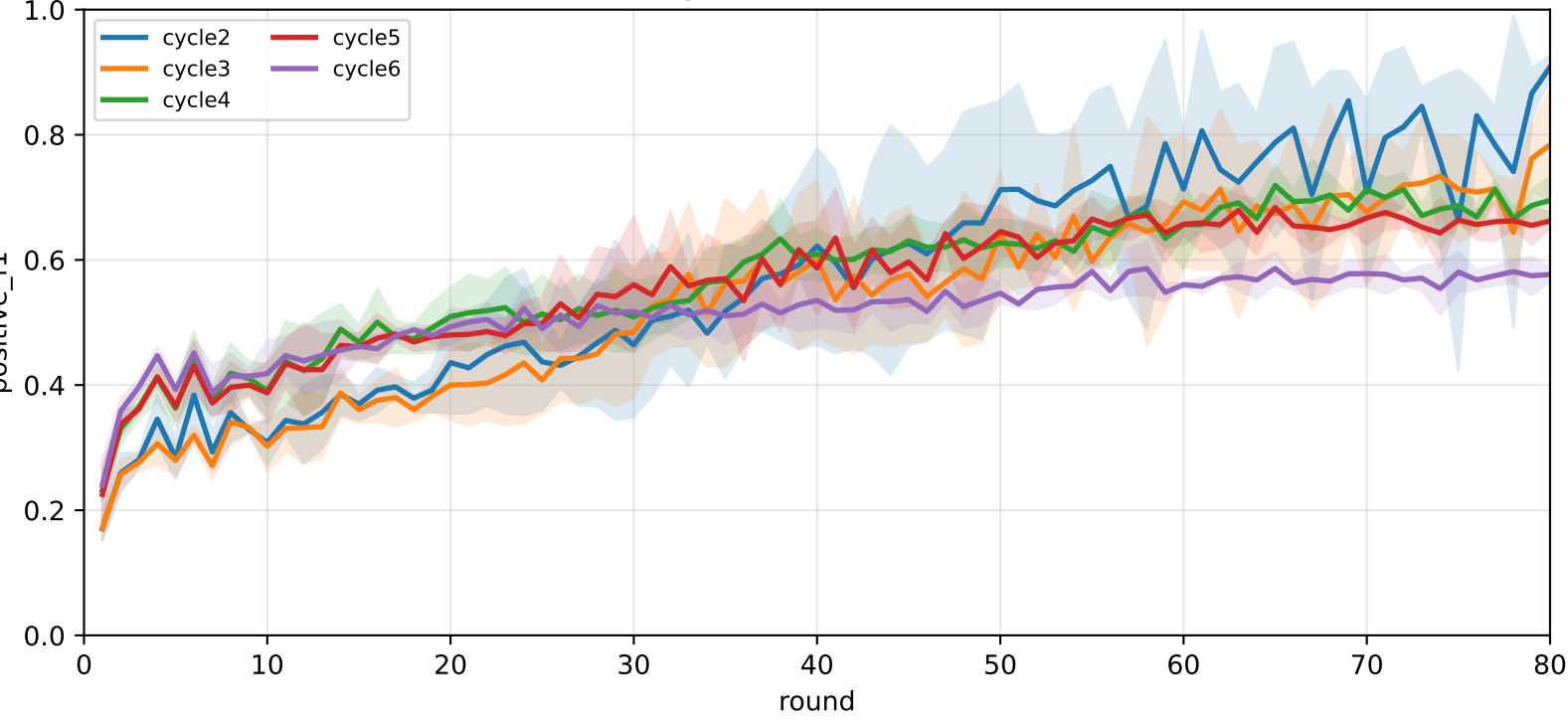
Pooled GCFL+ positive F1



Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)

